1 / Integrated drawing scope and source precedence

Coordinate contract: X right, Y toward operator, Z up; floor bottom Z0, floor top Z2.

Main case 600 x 320 x 20; six lower shells, six distinct flat panels and two additional roof halves.

Jogs fixed in Z at (110,165)/(490,165); D110 platter, no press action or shared-mode button yet.

Fader end assemblies translate outward 1.5 mm per side; retained axes and blade/caps do not move.

Roof is raised 5 mm relative to B24. Preserve 276 x 76 display reference and 272 x 72 opening.

U1/U2/U3/U4 mounting remains deliberately deferred under B25-ELEC-02; no guessed board holes.

Referenced PDFs are appended with historical labels; only selected F12 sensor pages are included.

This cover defines precedence only. Old DRAFT labels are historical and never confer new approval.

1. B26-PANELS-03 supersedes B26-PANELS-02's Y145 outlines; accepted F13-PANEL-06 sets Y165.

2. B26-REAR-POST-03 replaces four Y24 closure axes by Y8; five roof axes remain Y4.

3. B26-INTEGRATION-01 adds two rear C07s to B26-JOINTS-03: 16 bridges / 32 screws total.

4. F13-STATIC-02 modifies F13-STATIC-01: bottom-up fasteners, D50 sensor plate and ear pilots.

5. F13-STATIC-04 changes P02 feet top to local Z3.8 and extends webs down to Z3.8; cover stays Z4.

6. Only the listed F12-D05 sensor features apply: PCB pegs, toe bosses, nut entries, bores, wire trough and P08 toes. F12 spring reliefs, switch, push stroke, old housing and old platter do NOT apply.

7. F13-TOOLS-01 describes trial service tools only; no tool is part of the controller assembly.

8. K03 supersedes K02 only for CUE and PLAY/PAUSE logo size; K02/K01 retain other cap geometry.

2 / Assembly proposal, roles and outstanding checks

PRINTED: C01..C06 x1 each; six flat panels x1 each; D02L/D02R x1 each; C07 x16; 10 fader bridges; 30 switch retainers; 13 pot retainers; one encoder retainer; four display toes; 49 caps.

Two jogs each: P01 rotor/platter, P02 carrier, P04/P07 collars, P05/P06 covers, D50 sensor plate and two F12 P08 toes. These are proposed part quantities pending evaluated body/BOM verification.

PURCHASED/REFERENCE: bearings, fasteners, electronics, display, faders, potentiometers and switches.

Incomplete purchased bodies are labelled reference; they must not be exported as printed parts.

Legacy cap meshes are immutable display/reference assets, not editable feature history. Reconstruct or otherwise provide revision-matched editable cap sources before claiming the final source deliverable.

Nominal assembly intent: join lower shells; install controls and rear panels; assemble each jog off-case; approach from +Y with jog +6 Z as F13-PANEL-06; seat sensor/bosses and underside screws; fit front panels with caps removed; fit roof before R03 due to its front wedges; verify actual paths before use.

F13-TOOLS-01 uses nominal separated-prong trial interfaces; 0.45/0.5 mm radial features and 0.18 mm tool clearance are near FDM capability and NOT validated strength, calibrated fit or an operating tool.

There is no demonstrated complete 3D installed or moving assembly. All representations need BRep, full-head/nut, part/solid count, motion, distance, print-volume and viewer checks after the drawing gate.

Real display rear, purchased terminal/contact geometry, real wires and solder remain incomplete/reference.

Cables/plugs and mains protection are engineer-owned geometry-neutral assumptions, not a safety pass.

PETG intended; nominal printer 250 x 250 x 250 and nozzle0.4. Usable margins, layer, supports/brim, calibration, exact tolerance and physical acceptance thresholds remain open. Do not infer manufacturing readiness.

This candidate package is NOT an approved named complete drawing/model, STL, STEP or physical prototype.

3 / Immutable source snapshots

schematics/enclosure/B26-DETAIL-01/DJC_Enclosure_Rev_B26_DETAIL_01_drawing.pdf

SHA-256: 72b4371c01f78daf6964bd814d40c29256ee6c326642f7faccf73fe1bb7198e4

schematics/enclosure/B26-ROOF-01/DJC_Enclosure_Rev_B26_ROOF_01_drawing.pdf

SHA-256: cfefc4ac62c718c91429c0b8a43f91748fc49275d8140d93c959744605eb2576

schematics/enclosure/B26-REAR-POST-03/DJC_Enclosure_Rev_B26_REAR_POST_03_drawing.pdf

SHA-256: 4070fc141538226db621bfbed0045145544262d7487be7e3cfc82ca39c42adbb

schematics/enclosure/B26-JOINTS-03/DJC_Enclosure_Rev_B26_JOINTS_03_drawing.pdf

SHA-256: ba49faa2b4da4cfdeda69df188ef82cf3f8283d1bf23c56d586bbcc85d385319

schematics/enclosure/B26-INTEGRATION-01/B26_roof_shell_drawing.pdf

SHA-256: 17513176e93efbdde36ec6baf043920ecce6e5c4dfd3c29d7bcad57e731bc3ff

schematics/enclosure/B26-PANELS-03/DJC_Enclosure_Rev_B26_PANELS_03_drawing.pdf

SHA-256: bb18a44bedbeaa97a3462f3f8aab4248fdf212cd47080eed21cedaa086c2db8a

schematics/enclosure/B17/DJC_T_cap_Rev_B17_drawing.pdf

SHA-256: c9cc211f62c09b6274ae3f2a6a63a280fbe180e1c3fd1290c91369e0b444de71

schematics/frontpanel/K02/DJC_Keycaps_Rev_K02_drawing.pdf

SHA-256: 000de38e08a47f5edef9db8fac290c6f50a8996fb8c7c49af0b81cfc74e1097c

schematics/frontpanel/K03/DJC_Keycaps_Rev_K03_drawing.pdf

SHA-256: a31b600e5c6c5c7c2b0ef55fc7a9c4f03b409903709d1b4404a59faaf0638159

schematics/jog-wheel/F13-STATIC-01/Jog_Wheel_F13_STATIC_01_DRAFT.pdf

SHA-256: cf7f0fcaa0fe87bda85cc518b44de8ddc852f4cafb861fa54d747108774de6bf

schematics/jog-wheel/F13-STATIC-02/Jog_Wheel_F13_STATIC_02_drawing.pdf

SHA-256: 58fc5cb3ca1bd7416ddb8241e50e44cc302f6fc0962c3684bc7796761c8898c2

schematics/jog-wheel/F13-STATIC-04/Jog_Wheel_F13_STATIC_04_drawing.pdf

SHA-256: d95ba7c038bf770cabff6b38dea45f2806f51ff98137a92d7955c314043b1deb

4 / Immutable source snapshots

schematics/jog-wheel/F13-PANEL-06/F13_PANEL_06_drawing.pdf

SHA-256: 1a05ac331c32d99238d15ba1a164e33f5264d11e457b9624066925c8942269ef

schematics/jog-wheel/F13-TOOLS-01/F13_collar_tools_drawing.pdf

SHA-256: 58f3c3275f150c0b86c7a65154394cb86ea458800e187ac9d4df5ee1d382b371

schematics/jog-wheel/F12-D05/Jog_Wheel_Rev_F12_D05_DRAFT.pdf

SHA-256: a79451c414513671d489871711c969473be4e8ac8df0099339ff71c7c7e77938

schematics/frontpanel/R12/DJC_frontpanel_R12_DRAFT_coordinates.csv

SHA-256: 13a9926564f874d7f20e2aff74f90b460e42c0f264cfaa15fac10c6c1432559e

schematics/frontpanel/K01/DJC_Keycaps_Rev_K01_definition.json

SHA-256: 039bef206bfad70a354965938a142bc116d18959cbc7ef6983868ed3b914f649

schematics/frontpanel/K02/DJC_Keycaps_Rev_K02_definition.json

SHA-256: 38ca28baeefd6d4ca4296aedfca2f06bf8e9d266b823db5fd26d733a82634400

schematics/frontpanel/K02/DJC_Keycaps_Rev_K02_logo_sources.json

SHA-256: 790d5dce9a31d6a8a3c363181478f9dd61e89700863632bbb9f72707bb202845

schematics/frontpanel/K03/DJC_Keycaps_Rev_K03_definition.json

SHA-256: 17eb6b31df913ba0e59d658b9821d27b61ace95e66251596bfa5488f4886dcec

schematics/enclosure/B17/DJC_T_cap_Rev_B17_definition.json

SHA-256: d7d570cb13eab7d7a15df1f5c8ee629caa80bfdbfa5ba36b5b847444330d87ec

old_case/performance-pad.stl

SHA-256: 3ce7ede980638c21904d4a174e8813e1bc26c1091880e3aaca1b7772c3ebcf25

old_case/knob.stl

SHA-256: fa96654a5aa5c10a89864c4de4fb8a8e20842d8cc148bf176ed7efe8615998e4

old_case/tempo-fader.stl

SHA-256: 13162e64c43d38c626b1425eaa22c4e90ed775c208e042e82f436b194f785a6d

5 / Immutable source snapshots

old_case/crossfader.stl

SHA-256: c7d864b2908f8304e0f00e41a7e1111164eaf96696754c8835d2196cc82fec46

schematics/enclosure/B09/DJC_Enclosure_Rev_B09_controls.json

SHA-256: 4ce76301528235117bf29d7904b397543aa9666f941457c65a5edb41952bcb25

schematics/enclosure/B09/DJC_Enclosure_Rev_B09_equipment.json

SHA-256: 400fe43efc5237d0a3d1cd769444891d9eb679dffb15276c7e4c6e4c45b592da

schematics/enclosure/B19/DJC_Enclosure_Rev_B19_definition.json

SHA-256: 96665365ac4f9184dc25f814ae720b146a2cca46ed61890ed8653780fd720d5e

schematics/enclosure/B22/DJC_Enclosure_Rev_B22_definition.json

SHA-256: a01b52b87e15141ad86d5ca80f6a5a655a2df90d6f70022bc12143436875e27a

schematics/enclosure/B22/DJC_Enclosure_Rev_B22_validation.json

SHA-256: 8b6151816817865329b7290c28be702fdac38584b417d87889506a74ec2575bf

schematics/enclosure/B24/DJC_Enclosure_Rev_B24_validation.json

SHA-256: f2a7d273d8c8300eaab4db50a651f28ecbc0bce5deea65a7a6f7c23edb546b4a

schematics/enclosure/B24-JOINTS-02/DJC_Enclosure_Rev_B24_JOINTS_02_definition.json

SHA-256: 1488cd468959536bf8f2fd25393b111c007b28bc9dbf6ebd3669b504ae229b6d

schematics/enclosure/B24-AC01-LOW-01/DJC_Enclosure_Rev_B24_AC01_LOW_01_definition.json

SHA-256: b975e7955d3e0ae3108a664e6e48016757857c96595781a2f6de4c90366e59ef

schematics/enclosure/B25-ELEC-02/DJC_Enclosure_Rev_B25_ELEC_02_definition.json

SHA-256: 263a9abed777caa24dd7e4ba00fe1cc8fbf36f954ef19fa3cf31751a11c188d6

schematics/jog-wheel/F12-D05/Jog_Wheel_Rev_F12_D05_definition.json

SHA-256: 0798e7ca6d62e07e768f4e262c47525c98effcb4b56a275a5da85e81228f9e7d

models/enclosure/B15/DJC_Enclosure_Rev_B15_inputs.scad

SHA-256: 8f2bd047b6fb1a6002a2f386a0a6f6fa6f13e3d72c7844b5e02de7f2349dde28

6 / Immutable source snapshots

models/enclosure/B19/FUSION/DJC_Enclosure_Rev_B19_Fusion01_BOM_and_exports.json

SHA-256: 30df63546706003336 29c89e68233ec1090a4d4f66afb657bbb24421a6df7451

models/enclosure/B24/FUSION/DJC_Enclosure_Rev_B24_Fusion05_BOM_and_exports.json

SHA-256: da981aa40ceef06053b538f24089e7b39331c15463f023f8e38ad9c16272a2f4

models/enclosure/B24/FUSION/DJC_Enclosure_Rev_B24_Fusion05_validation.json

SHA-256: 8c8763c0427f119f73fb7061c340cf74baaa1c86386e6f1529abfe795974d5ae

models/enclosure/B16/FUSION/CAP_B_Rev_B16_Fusion01.stl

SHA-256: 37d5ff82b539e87fa27ffa795a6c2493bcc57afe4a1ff978b7a8fa7b99d8d87e

models/enclosure/B17/FUSION/CAP_T_Rev_B17_Fusion01.stl

SHA-256: 8e0d93ec5d0d7b290face7c6e1e3f0654b98c07a7d0edf9917f895f48957b858

models/enclosure/B16/FUSION/CAP_P_Rev_B16_Fusion01.stl

SHA-256: 6bc7a516eeb4ba382fe5be5cce5e460289db7dfdfdd1eea1099107b9478309e3

models/enclosure/B16/FUSION/CAP_F_Rev_B16_Fusion01.stl

SHA-256: 5a92948fe01db726faa5b6867d5787b951b299b783bebac09769aca1e67bd7a9

models/enclosure/B16/FUSION/CAP_C_Rev_B16_Fusion01.stl

SHA-256: 0926fcbc70240fd979c26ddca384f61f02ab50440658e5212a06eacf41995a99

models/enclosure/B16/FUSION/CAP_E_Rev_B16_Fusion01.stl

SHA-256: cb5f72be8ceb9e4cdcee00165a97c2bc24370b56046e7c9e211207f4000861d2

models/enclosure/B24/FUSION/KEYCAPS_K03/CAP-CUE40_Rev_K03_Fusion03.stl

SHA-256: b5f8374b7fe40e5e8d02d47cfbf4d6c739d020cb72bfb5ba36059ce4b38ce571

models/enclosure/B24/FUSION/KEYCAPS_K03/CAP-PP40_Rev_K03_Fusion03.stl

SHA-256: 59245a23c144f44d41961fe8b7d75fd3b78d0cc92ffb29293b260076498d332d

models/enclosure/B24/FUSION/KEYCAPS_K03/CAP-HC40_Rev_K03_Fusion03.stl

SHA-256: a7d5ed2b3b24f522ea4d68b192ed64052bf3c69c7bc3b1e5eae24732725e0e31

7 / Immutable source snapshots

models/enclosure/B24/BUILD123D/shell.py

SHA-256: 0d11cdf8a3aeae5177434d5060853466001942a1022ba6f292a0336813bf747f

models/enclosure/B24/BUILD123D/panels.py

SHA-256: 83b70993090b0274be37c26aa376f47db04fbbbde9c7762685b7e3f337a674d6

models/enclosure/B24/BUILD123D/controls.py

SHA-256: 88d26d78b9f2de7f59645a09be26a27fa5d5fb21027a060a2115fe835243f0cd

models/enclosure/B24/BUILD123D/caps.py

SHA-256: c9b9be49f4d75dd0129c263533ece12244a76d0c74aa6aadfe1f89e042457357

models/enclosure/B24/BUILD123D/bridges.py

SHA-256: 6e1b43bdbc33dab9b07de6ec4eab651ff51c87f8cce9ef4c7c0ee0e9f252a5b1

models/enclosure/B24/BUILD123D/hardware.py

SHA-256: 7fae4ee5e1f0a71dce40ed9ec69c24f28473f73e4ac025fd0dda8bf7a8a65a89

models/enclosure/B24/BUILD123D/display_toes.py

SHA-256: b5336e8679abe28127d365fb7bdf13b411d1c3d62e1a98bd8a47d46b7b2f8102

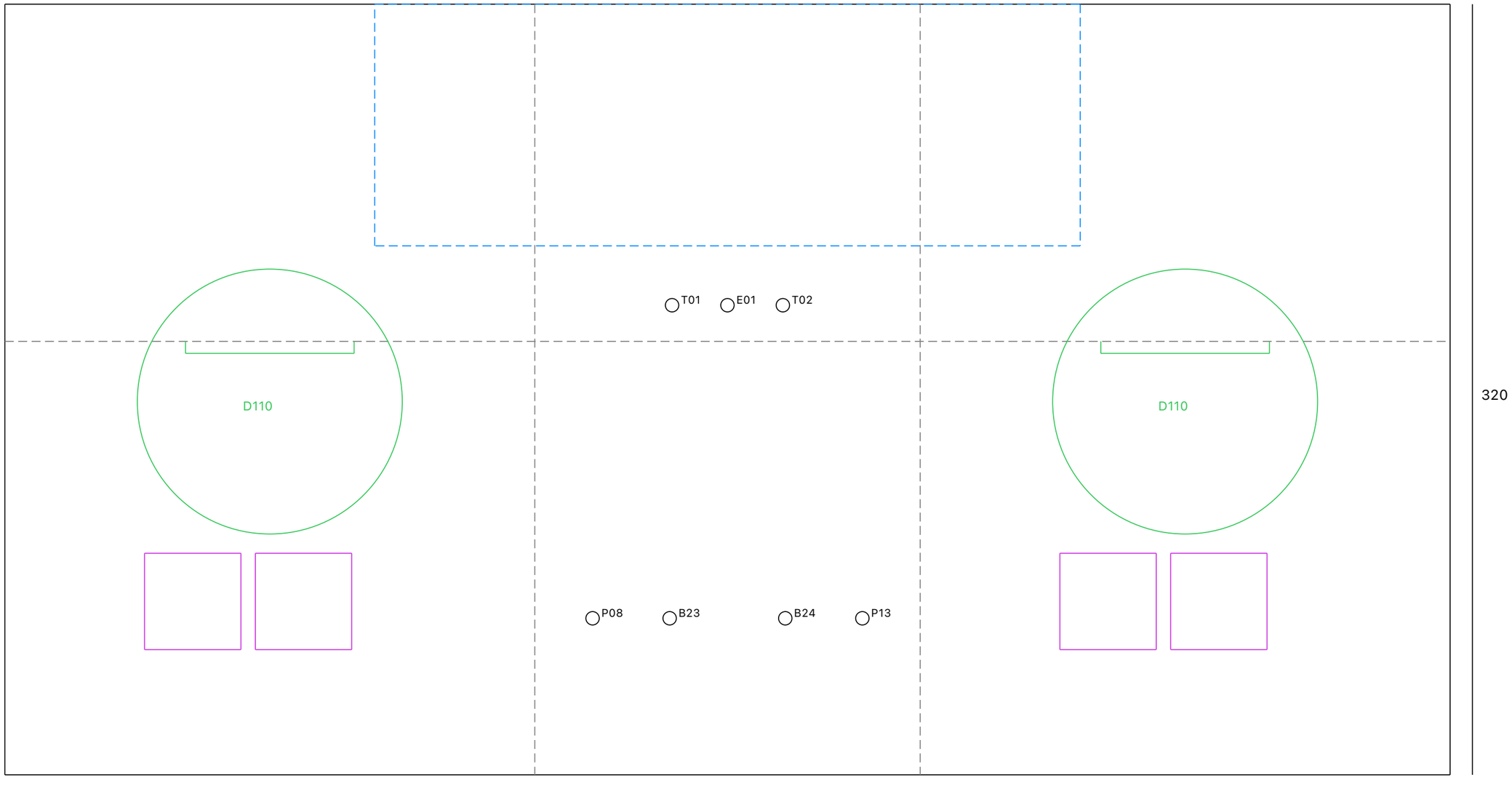
models/enclosure/B24/BUILD123D/reference_controls.py

SHA-256: eb3d81227f8328b122c5e0fb5d3b1e1c20ea9f61364c8310733cf44fdf7343a7

models/enclosure/B24/BUILD123D/reference_equipment.py

SHA-256: ed5748be9ea3f03f2eeff4313de60336b42c244489b28c2d7dff1fdc10b10f75

1 / Datum plan and changed interfaces



mm; origin = NEW rear-left, X right, Y toward operator, Z up; plan diagram, not cutting profiles.

Old fixed control coordinates -> (X,Y+20). Exceptions E01/T01/T02 and PFL per table on sheet 2.

Jogs old (110,150)/(490,150) -> (110,165)/(490,165): -5 relative to fixed controls.

K03 squares: B13/B14/B19/B20 centres Y248, 40 x 40 max; nearest B14/B20 gap 8.0 nominal.

sqrt(max(14-20,0)^2 + (248-20-165)^2) - 55 = 8.0 mm; no fit or motion qualification.

Grey dashed: shell seam; green: proposed stepped panel seams. Neither is yet validated.

Additional controls and all screw heads omitted from picture: reconstruct from the frozen sources, not this diagram.

2 / Exact delta, control and print-part schedule

A. BASELINE: B24-build123d01 source; B26-STUDY-01 is accepted partial layout only.

Main flat body envelope 600 x 320 x 20. Old Y0..300 becomes Y20..320; X/Z unchanged.

Shell X splits 220,380; Y split 120 -> 140. Six shells; rear strip see B26-ROOF-01 sheet 4.

Flat panels R01/R03/R04 and R05/R06/R07: old regions from B19 + B22/panels.py, Y+20.

Rear edge expansion and display subtraction change their contours; detailed joints/edges remain TBD.

Roof halves D02L X153.55..335 and D02R X335..446.45 are TWO additional separate parts.

They are NOT counted in six flat panels. Roof local shape moves rear 20 and up 5 relative to body.

New datum: screen X/Y unchanged, Z+5 from B22; roof front junction Y100.346; angle 49.

Roof seat centre (300,69.757639,48.188383) -> (300,69.757639,53.188383).

Rear upper Z80.376766 -> 85.376766. Candidate junction Y100.346: see B26-ROOF-01.

E01 (300,125), T01 (277,125), T02 (323,125); T01/T02 actual B09 head, not 10 mm boxes.

B23/B24 (276,255)/(324,255), aligned with P08/P13 (244,255)/(356,255).

B01..B10 and P01..P03 use B24 moved X from B26-STUDY-01 line 18, their R12 Y +20.

All other control axes including F01-F05 use R12 coordinates translated Y+20; no toggle hole.

F01 (196,272), F02 (576,272), F03 (276,197), F04 (324,197), F05 (300,301).

NOMINAL PRINT BOUNDS, before protrusions, joints, roof interfaces and corner radii:

Shell front row: 220 x 180, 160 x 180, 220 x 180; rear row: 220 x 140, 160 x 140, 220 x 140.

Flat-panel nominal max Y spans: rear outer 145, centre R03 39.654; front row 180.

Candidate R03 X220..380, Y100.346434..140; edges, cutouts and steps require integration.

Roof split widths 181.45 / 111.45; left lap adds 10. Full print orientation and protrusions TBD.

Nominal 250 mm cube and 0.4 nozzle only: no assertion of usable bed, fit, supports or strength.

F13-STATIC-02 adds a proposed bottom-access jog mount; complete integration remains open.

Legacy panel D108 well is smaller than platter D110 in plan; fit/appearance require review.

D52 aperture, sensor, fixed carrier, bearing seats, hardware and mounting are not finalized here.

F12 local bottom 21.4, top 33.4 -> top 36.4; local-to-case Z registration remains to be verified.

No axial pressing, springs or jog switches; retain AS5048A and D8 x 4 diametric magnet.

The nominal bounds do not establish a complete model: see required interface closures on sheet 4.

3 / Fader end cup and bridge: exact translated delta

B09 fader_family is inherited in local (u,v,z): u travel, v transverse, z=Z-S.

F01-F04: (X,Y)=(X0+v,Y0+u); F05: (X,Y)=(X0+u,Y0-v). Seat S retains source.

Move EACH complete end assembly outward 1.5 along u, including end wall, end wings, tower,

pilot, removable end bridge, through bore and countersink; leave central rails/body/slot/cap fixed.

Implement as rigid u translation of each B09 end-side primitive and both bridges, not scaling L.

B09 additions lines 142-148, cuts 150, bridge 154-155: use u'=u+sign(u)*1.5.

Old tower axes +/-31.25 (F01-F04), +/-31.10 (F05); new +/-32.75, +/-32.60.

Tower radius 2.2; nearest-to-body axial gap 32.75-2.2-30=0.55 (F01-F04).

F05 gap 32.60-2.2-30=0.40. These are proxy gaps to nominal 60-long box, not fit claims.

End cup inner faces move from +/-30.25 to +/-31.75, and +/-30.10 to +/-31.60.

Central bearing rails remain u=-15..15; end wings now occupy abs(u)=24.75..31.75

for F01-F04, 24.60..31.60 for F05. Connectivity to rails and floor remains subject to 3D review.

Bridge inner edge moves from abs(u)=27.25/27.10 to 28.75/28.60.

Nominal overlap with each 60 mm body end becomes 1.25/1.40; verify real restraint, lift and tolerance.

M2 countersunk max head D4.4, bore D2.2, pilot D1.6 and seat Z inherited from B09; qualify hardware.

Absolute tower axes (new rear-left datum; each pair shown on u):

F01 (196,239.25)/(196,304.75); F02 (576,239.25)/(576,304.75).

F03 (276,164.25)/(276,229.75); F04 (324,164.25)/(324,229.75).

F05 (267.40,301)/(332.60,301). All axes Z and screw engagements from frozen model/B09.

F01/F02 lower tower head ends at Y306.95 nominal. Case/front-wall and driver path not checked.

F03/F04 front tower head centres Y229.75: PFL centres Y255; actual caps/head access TBD.

F01/F02 jog and fader assembly interact with bowl, wall, retained bridge and surrounding supports:

full 3D material connectivity, end-cup contact, screw-head/tool sweep and assembly order TBD.

Real fader terminal/solder protrusions, exact body centering and measurement tolerances TBD.

This delta is an engineer-review proposal, not proof of assembly or mechanical strength.

4 / Sections, source hierarchy and blocking gaps

SECTION A-A / see B26-ROOF-01: candidate roof angle49, raised 5 with front junction Y100.346.

Flat rear upper Z85.376766 for Y0..43.515; 2 mm vertical section retained.

Opening remains 272 x 72; two half-roofs remain, not a complete real vendor body.

Front/rear flange/posts: see B26-ROOF-01 sheets 3/4; complete driver access remains TBD.

SECTION B-B / flat body: floor Z0..2, panel Z18..20, old wall 2; rear extension joining TBD.

B24 well D108, floor Z19: D110 exceeds it in plan; no collision is implied by diameter alone.

B24 joints: C07 28x10x3, hole axes +/-9, D3.4; screws head D5.5 x 3, Z11..14.

Translate old joint layout +20 in Y only as reference; joints at new rear strip require design.

Panel countersink nominal D3.4 to D6.0, depth 1.3; actual screw-head fit still TBD.

33 source posts, including 7 proposed relocations, plus two roof lap axes need full-head checks.

Nominal split bounds exclude C07 protrusion, local cutouts and assembly tooling: not bed validation.

B25-ELEC-02 board placements approved as placement drawing only, not implemented in B24 model.

Transform placement references +20 Y: U1 (28.75,170), U2 (181,185), U3 (419,170),

U4 (571.25,170); actual U4 header/terminal outline unknown. No board mounts are specified.

Recheck boards, complete bodies and connector/service/wiring access against moved jog and seams.

SOURCE MAP / reconstruct unchanged geometry from frozen sources, not graphics:

B26-STUDY-01 drawing.swift lines 16-18, 46-90, 92-115; partial accepted layout only.

B24 BUILD123D shell.py and panels.py, B15 inputs.scad, B19 definition panel_regions/fader_recesses;

B22 definition main_body/display_housing; B09 controls fader_family and button_family;

R12 coordinates CSV; B24 definition control_moves; B24-JOINTS-02 definition; B25-ELEC-02 definition.

BLOCKERS: complete display rear/connector/body; mux header; bought terminals and screw head profiles;

roof fixings; panel seam steps; jog recess and carrier; joint interfaces; tool and service access.

B09 triangle actual 18.2-wide head and rotation, switch retainers and complete caps must be checked

against screen transition, joints and screw heads. R12 10x10 is only a reference, not an installed cap.

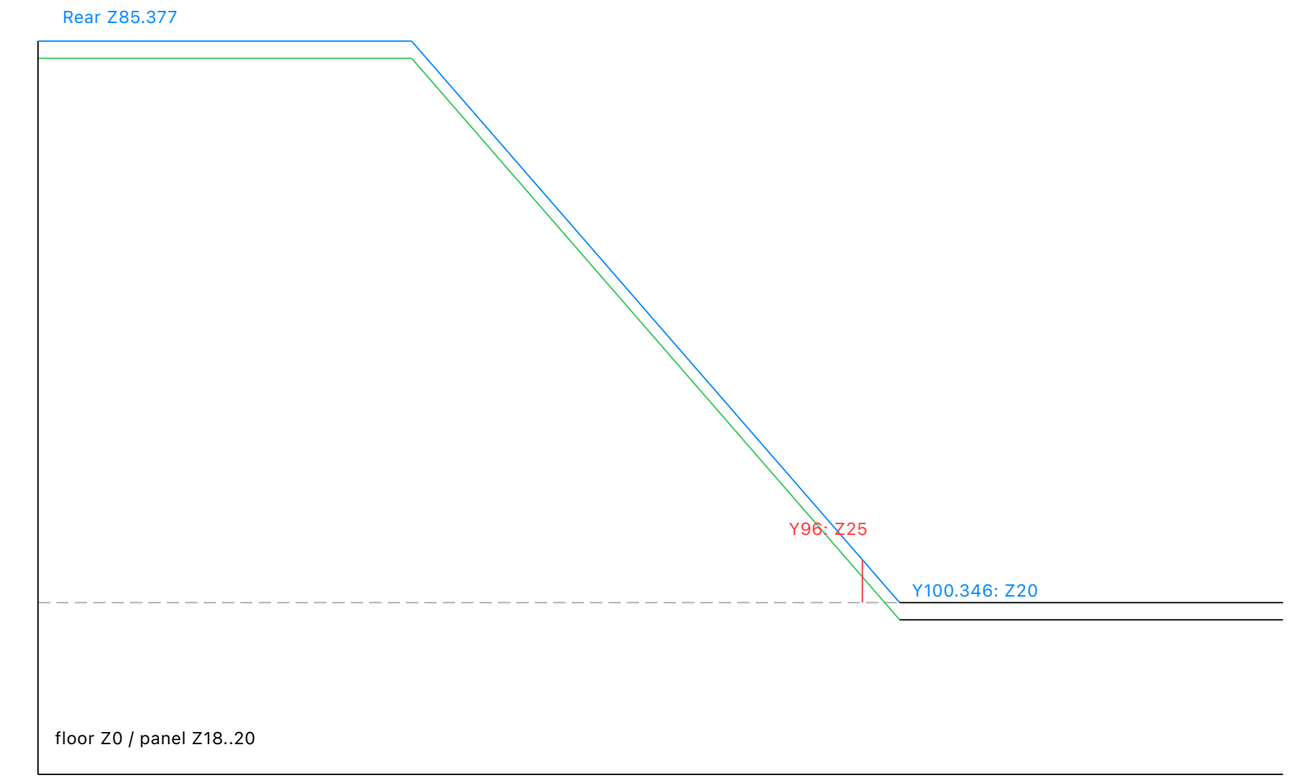
PETG fit, shrinkage, tolerances, bed margins, orientations, support, strength and safety all TBD.

No electrical, mains, thermal or energizing release; no collision, physical or manufacturing approval.

Do not make 3D geometry until responsible engineer approves a named complete drawing.

1 / Candidate continuous raised roof

SIDE SECTION / 1.4:1 / not a complete solid or assembly



Reference: B22 roof angle 49 deg, original junction Y96/Z20.

Retain the same roof orientation and move the screen/roof 5 up.

On the moved slope: U(Y)=25+tan(49 deg)*(96-Y).

Extend the sloping material until it meets the main panel Z20:

Yj = 96 + 5/tan(49 deg) = 100.346434.

At Y96 upper is 25; at Yj upper is 20: no added front wall.

Below each upper datum use the inherited 2 vertical mm wall.

Rear horizontal top: 85.376766, breakpoint Y43.515278.

The screen opening remains 272 x 72 in its tilted plane;

display is not stretched. Screen rear and complete connectors

are not dimensioned; this section does not depict those bodies.

This is a calculated face extension, not a tested solid, joint or complete display installation.

Old-to-new table origin translation adds Y20 to unmoved controls. Screen itself moves rear 20

relative to controls, so its new table Y is numerically unchanged; its global Z increases 5.

Body exterior 600 x 320 x 20 is a proposal; the display roof rises above the 20 mm main body.

2 / Layout derivations and remaining interfaces

Basis: accepted B26-STUDY-01 is a partial layout, not complete drawing approval.

Use B22 display-housing local face and B26-DETAIL-01 translation as the unchanged basis.

Outer X envelope 153.55..446.45; existing roof split X335 and two roof components retained.

Screen seat centre (300,69.7576388404,48.1883832089) -> (300,69.7576388404,53.1883832089).

Rear horizontal-to-slope break: Y=96-80*cos(49)=43.515278, upper Z85.376766.

Candidate slope interval Y43.515278..100.346434; U=25+tan(49)*(96-Y), lower U-2.

Continuity: U(43.515278)=85.376766; U(100.346434)=20, lower=18.

Triangle controls T01/T02 at Y125 have conservative head rear envelope Y=125-9.1=115.9.

Gap between candidate front junction and that envelope: 115.9-100.346434 = 15.553566.

This is a planar envelope result only: it excludes actual stems, retainers, bolts and service tools.

JA/JB at (110,165)/(490,165) remain separate jog references; no all-body collision claim.

Do not cut the installed display into pieces. The six flat panels exclude two roof halves.

The moved screen contact and roof toes/pockets are inherited but need actual frame-land verification.

Candidate front flange/posts are on sheet 3; rear posts and wall are proposed on sheet 4.

Check complete fastener heads, fader towers, split joints, plate insertion, electronics and real display rear.

The old B22 module front retention positions and all purchased connectors are not assumed valid here.

R03 and side-panel mating cuts must follow the new junction; their new outlines are not yet defined.

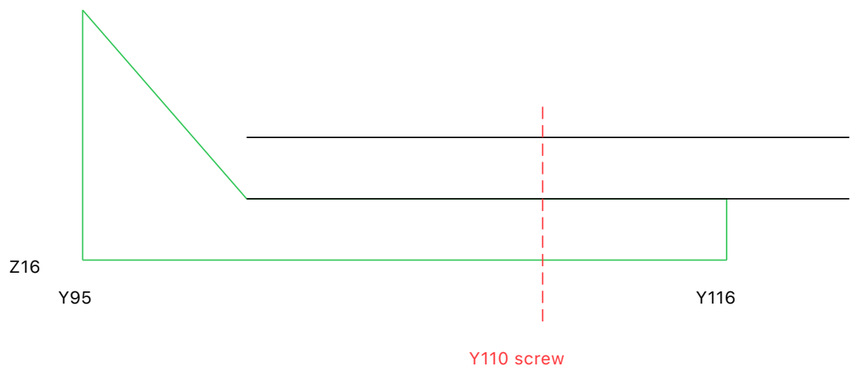
No CNC/print release, connected wiring, mains/thermal qualification or physical-fit approval is implied.

OPEN: actual rear-body fit / seams / installed-body coverage / print orientation and margins.

This face solution alone does not establish a complete reconstructible B26 drawing.

3 / Candidate front flange and two mounting posts

SECTION THROUGH FLANGE / 5:1 / nominal; full assembly not shown



Two candidate integral flanges, one per roof half:
X223..233 and X367..377. Extrude the SAME Y/Z polygon:
(95,16), (116,16), (116,18), (100.346434,18),
(95,24.150368), close. Upper sloping edge matches roof
underside U(Y)-2; join along the full sloped face.
B22 retained front flanges used Y95..116 / Z16..18.
This proposal adds only the wedge needed for raised roof.
Cut D3.4 through flange Z16..18 at (228,110)/(372,110).
Roof/panel fuse/contact and exact fillets require later validation.

POSTS, PANEL AND FASTENERS / PROPOSAL

Move the two front floor posts to (228,110)/(372,110) in the new datum. Each source-type post: D8 core Z2..16, D9 foot Z2..5.8, pilot D2.5 Z6..16. Actual printed thread needs tests.
Extend R03 rear edge to Y100.346434, retain front edge Y140 and X220..380, panel Z18..20.
At each post axis: R03 through D3.4 and 90-deg countersink D6 / 1.3 deep, as B24 source.
Retain nominal Bossard BN616 M3x8 length including head: top Z20, tip Z12; flange clears Z16..18.
Nominal pilot engagement Z12..16 = 4; tip-to-pilot-bottom Z12-6 = 6. This does not certify grip.
A post at old screen-axis Y112 would touch the J01/J04 bridge bounds at Y116 in plan.
With Y110, D8 core edge Y114 gives 2 mm nominal plan gap to bridge Y116. Nearest D9 pad at (229,121)/(371,121) is 2.545 mm clear from the post core in plan. Only these simple references were compared; surrounding solid shapes, full heads and tool access are unverified.
Rear post and roof-lap proposal is defined on sheet 4; actual display and service remain open.

NO all-body collision, print strength, actual screw fit or assembled service proof.

4 / Rear wall, side returns, posts and lap

Use new rear-left table datum; main floor Z0..2, rear wall thickness 2.

For each rear shell bay, retain B24 wall at Y0..2 to Z18. Above Z18 add a raised wall

only at X153.55..446.45, ending Z83.376766; split at shell planes X220 and X380.

Remove the translated former rear-wall material at Y20..22 only as the original base

wall feature, before reattaching separately defined retained controls and equipment.

Keep the source-derived rear openings on the NEW wall Y0..2, not on the former wall:

RCA full notch X37.25..72.15, Z2..18; D3.6 axes X32.75/76.65, Z9.95.

AC01 cutout X157.025..204.525, Z3.817176..31.817176, from B24-AC01-LOW-01.

48 ventilation cells: reuse exact B22 validation polygons in X/Z, on Y0..2, without scaling.

RCA plug/solder, AC01 clips, terminal guards, actual connected parts and thermal safety are not modeled.

Roof side returns: X153.55..155.55 and X444.45..446.45. Extrude each Y/Z polygon

(2,18), (100.346434,18), (100.346434,20), (43.515278,85.376766),

(2,85.376766), close. Retain 2 mm side return width; join roof material, then cut opening.

Move the 5 rear post axes to Y4 at X210/230/300/370/390; do not retain the old translated axes.

Each post: D8 core Z2..83.376766, source D9-to-D8 foot Z2..5.8, pilot D2.5

Z73.376766..83.376766. Roof opening D3.4 and 90-deg countersink D6 / depth 1.3.

Nominal retained BN616 M3x8 top Z85.376766; tip Z77.376766; post engagement 6, reserve 4.

Retain source left-roof lap X325..345, Y13..29, now Z75.376766..83.376766.

Right roof through D3.4 / countersink at (340,18)/(340,24); left lap pilot D2.5

Z76.376766..83.376766. Screw tip Z77.376766 gives nominal engagement 6 and reserve 1.

Carry screen-face frame, supports, four toe interfaces and their fasteners by the selected Z+5

relative to B24 source, with actual display rear, lands, connector and hand clearance unqualified.

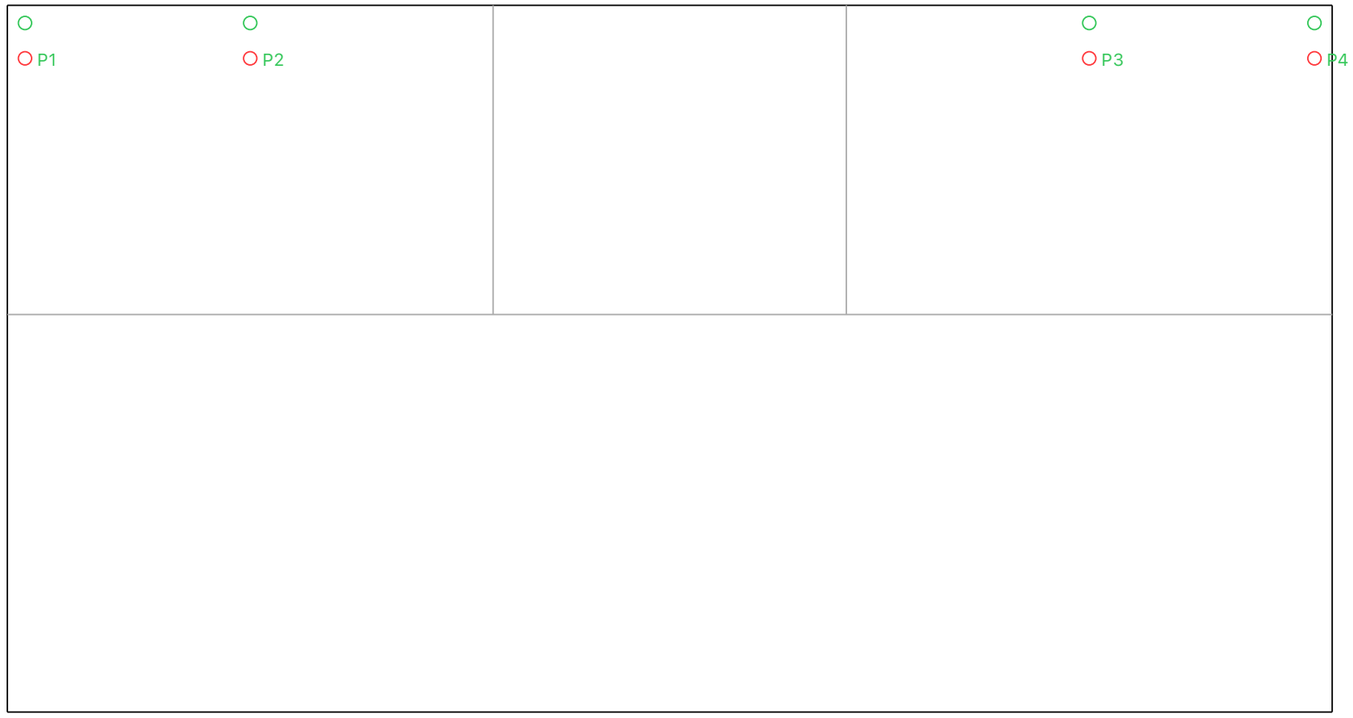
Keep the five posts, two lap holes, source lower wall openings and their counts distinct.

No rear-body collision, head-bearing, fit, wall strength, electrical or thermal certification.

TBD: floor-to-wall joint integrity, true installed connectors, full tool sweep and print settings.

1 / Relocate four outer rear panel closures

PLAN / new rear-left datum / the rectangle is the complete 600 x 320 body



Green: proposed Y8; red: former Y24

Retain X = 8, 110, 490, 592. Move the four outer rear panel post and hole axes from Y24 to Y8.

P1/P2 belong to R01/C01; P3/P4 belong to R04/C03. Delete only their former Y24 posts and holes.

The five tall roof posts at X210/230/300/370/390, Y4 are unchanged and are not panel posts.

Reason: Y8 matches the retained front-panel 8 mm edge axis rather than leaving the extended rear

edge 24 mm from the nearest original screw. This is a nominal support proposal, not a stiffness proof.

No bought part, jog, control axis, fader, display opening or six-panel boundary is moved.

Prior B26 approved drawings remain unchanged. This dimension change requires named approval.

2 / Retained post section and scoped proximity checks

Per axis: integral post core D8, case Z2..18; D9-to-D8 foot Z2..5.8, source B24 geometry.

Pilot D2.5, Z8..18; panel Z18..20, D3.4 through and D6/90deg countersink from Z20, depth 1.3.

Retain BN616 M3x8 as a nominal reference; screw fit, thread formation and load remain unqualified.

Floor Z0..2; join each post foot to its owning C01 or C03 floor. Wall at Y0..2 stays unchanged.

At Y8 the nominal D9 foot starts at Y3.5: 1.5 mm from that wall; the floor is its support path.

The nominal D6 countersink edge lies 5 mm from the rear edge; subtracting the source R0.8 outer

edge fillet leaves 4.2 mm in plan. Corner fillets, real screw heads and physical fit still need checking.

LIMITED NUMERICAL SCREEN / plan bounds only

4 proposed D6 heads vs 49 control-aperture bounds, expanded 0.8 mm: 196 pairs, minimum 22.950 mm.

4 D8 cores vs 14 translated C07 plate rectangles: 56 pairs, minimum 119.065 mm.

4 D9 foot circles vs 28 D9 joint-pad circles: 112 pairs, minimum 120.973 mm.

4 D9 feet vs the RCA and AC01 wall-opening plan bounds: 8 pairs, minimum 25.359 mm.

Computed bounds do not cover actual port bodies, wires, other mounts, tools, roof/body sweeps or forces.

Prior F13-STATIC-04 approved 0.2 mm foot clearance is unrelated and remains unchanged.

OPEN / NOT A COMPLETE RECONSTRUCTIBLE B26 DRAWING

Assess real port/connector/hardware envelopes, post/floor integrity, panel edge and D6 countersink.

Actual 3D interference, print margins, fader supports, full assembly/service and strength remain open.

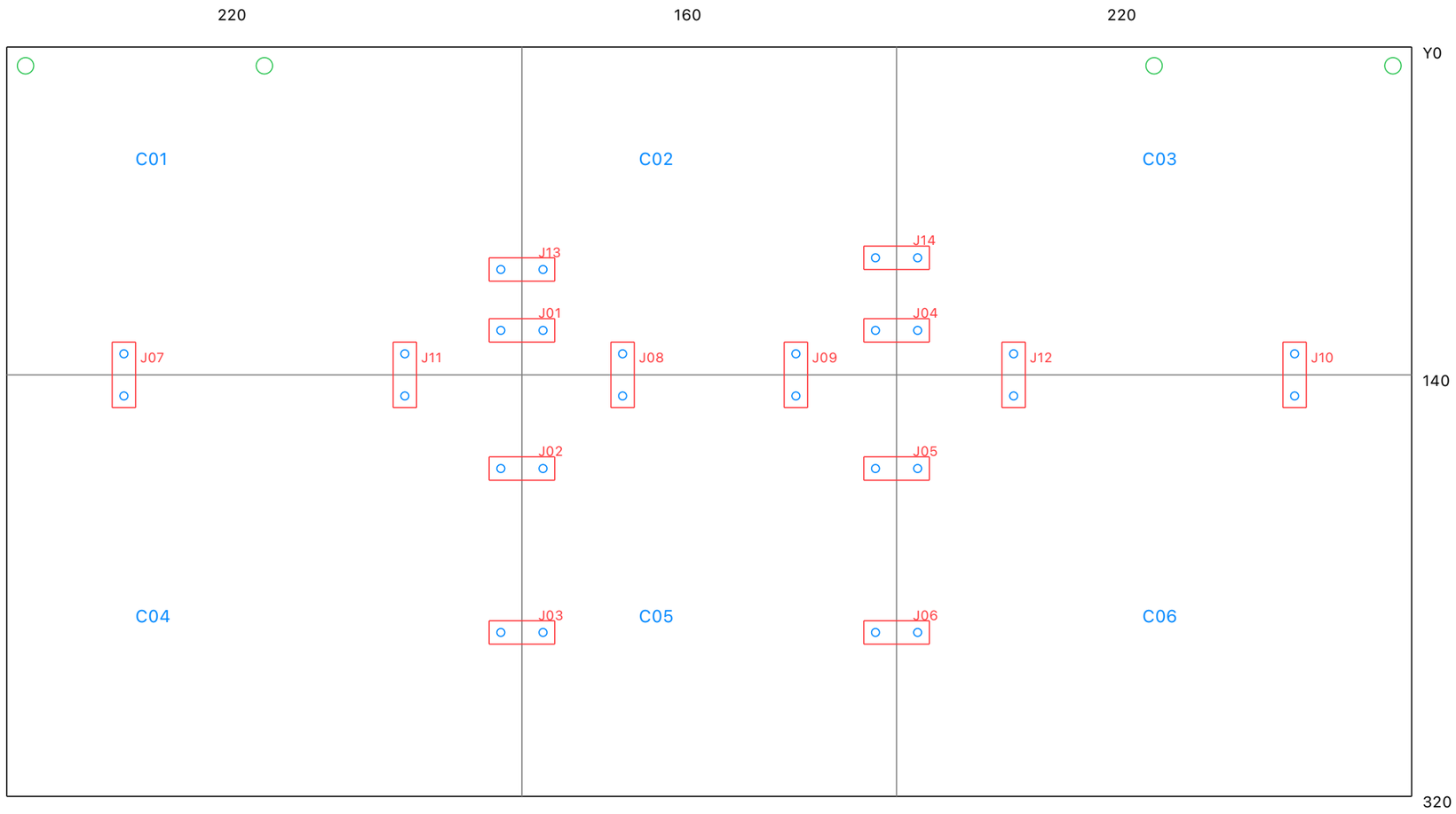
A screen/roof-to-panel interface and all shell seam transitions still require integrated definition.

No 3D model or physical fit check was run. New B26-REAR-POST-03 has not been approved.

Sources: B26-PANELS-02 pp1-3; B24-JOINTS-02; B22 post source; R12 CSV; B26-STUDY-01; B26-ROOF-01.

1 / Proposed six-shell partition and transferred joints

PLAN / nominal shell domains and bridge rectangles; all geometry candidate except named rear axes



Green circles: approved B26-REAR-POST-03 rear closure axes, NOT C07 joint axes: (8,8), (110,8), (490,8), (592,8).

Five roof posts at Y4 remain separate.

Red rectangles: C07 candidate transfer.

Blue circles: 2 x D3.4 bridge holes.

Labels refer to schedule on sheet 2.

Shell ranges: C01 [0,220] x [0,140]; C02 [220,380] x [0,140]; C03 [380,600] x [0,140].

C04 [0,220] x [140,320]; C05 [220,380] x [140,320]; C06 [380,600] x [140,320].

Floor nominal Z0..2. Seams X220/X380 and Y140 are proposed butt boundaries, not qualified interfaces.

Rear strip Y0..20 extends the three rear shells. Joins at X220/X380 in that strip have no local fastener in this transfer; wall, floor continuity, roof mating and any additional connection remain TBD.

Source B26-DETAIL-01 sheet 2; B24-JOINTS-02 definition. Transfer X unchanged, Y +20, Z unchanged.

2 / Joint and integral pad ownership schedule

Each row: centre and two pad/pilot axes in global XY, mm. A = negative normal; B = positive normal.

ID	N	centre X,Y	A: owner / pad axis	B: owner / pad axis
J01	X	(220,121)	C01 (211,121)	C02 (229,121)
J02	X	(220,180)	C04 (211,180)	C05 (229,180)
J03	X	(220,250)	C04 (211,250)	C05 (229,250)
J04	X	(380,121)	C02 (371,121)	C03 (389,121)
J05	X	(380,180)	C05 (371,180)	C06 (389,180)
J06	X	(380,250)	C05 (371,250)	C06 (389,250)
J07	Y	(50,140)	C01 (50,131)	C04 (50,149)
J08	Y	(263,140)	C02 (263,131)	C05 (263,149)
J09	Y	(337,140)	C02 (337,131)	C05 (337,149)
J10	Y	(550,140)	C03 (550,131)	C06 (550,149)
J11	Y	(170,140)	C01 (170,131)	C04 (170,149)
J12	Y	(430,140)	C03 (430,131)	C06 (430,149)
J13	X	(220, 95)	C01 (211, 95)	C02 (229, 95)
J14	X	(380, 90)	C02 (371, 90)	C03 (389, 90)

Fourteen plates, 28 integral pads, 28 screws; seven adjacent shell pairs have two bridges each.

C01: 4 pads; C02: 6; C03: 4; C04: 4; C05: 6; C06: 4. Pads are shell features, not loose parts.

A and B identify owning shell on negative / positive X or Y side of the relevant seam respectively.

Plate local u is across seam: +u is +X for N=X, +Y for N=Y; v is along seam; w is global Z.

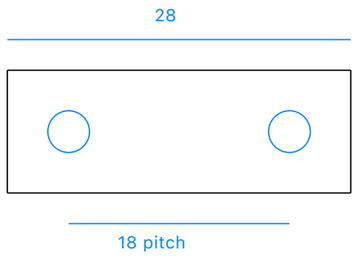
Centres and axes are transferred candidates, not newly approved B26 dimensions or a collision result.

Cited frozen source: B24-JOINTS-02 definition, retained_joints + added_joints; approval: docs/APPROVALS.md.

B26-PANELS-02 sheet 5 proposes this same X unchanged / Y+20 transfer; it remains unapproved.

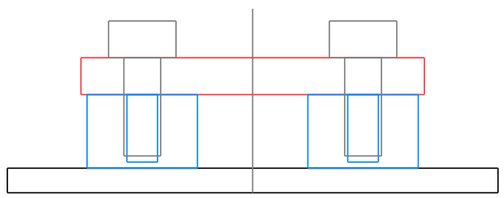
3 / C07 detail, assembly intent and integration limits

C07 PLAN / local u across seam; nominal 2:1



10 wide; 2 x D3.4 through

SECTION A-A / along u through both axes; nominal 2:1



global Z14 screw head top

Z11 head bottom / plate top

Z8 plate bottom / pad top

Z2 floor top / pad base

Z0 floor underside datum

Local w = global Z, not relative to panel or jog. C07: 28 x 10 x 3, u=-14..14, v=-5..5, Z8..11.

Holes at u=-9,+9, v=0; D9 integral pads Z2..8; blind D2.5 pilot Z2.5..8, depth 5.5.

M3x8 ISO4762 reference: shaft Z3..11, D5.5 x 3 head Z11..14; nominal pilot engagement 5, tip reserve 0.5.

Printed: C01-C06 x1 each, C07 x14; 28 pads integral. Purchased: joint M3x8 socket-head screws x28.

Hardware envelope is a source reference, not measured hardware; no thread or socket detail implied.

Intent: join six lower shells with 14 bridges from above before equipment, roof and panels. Fixture, driver sweep, actual order, access and connected-service removal are TBD; do not assume an executable assembly.

B26-PANELS-02 sheet 5 reports XY bounds ONLY: 700 plate/fader pairs min 3.300; 1400 pad/fader pairs min 4.244; 98 selected roof-core/plate pairs min 2.000; 196 roof-foot/pad pairs min 2.045 mm.

Those source screens exclude real bodies, tools, cables, seams and all complete 3D overlaps; not fit proof.

Roof joins and rear strip connections, electronic bodies, fastener interference and jog service remain open.

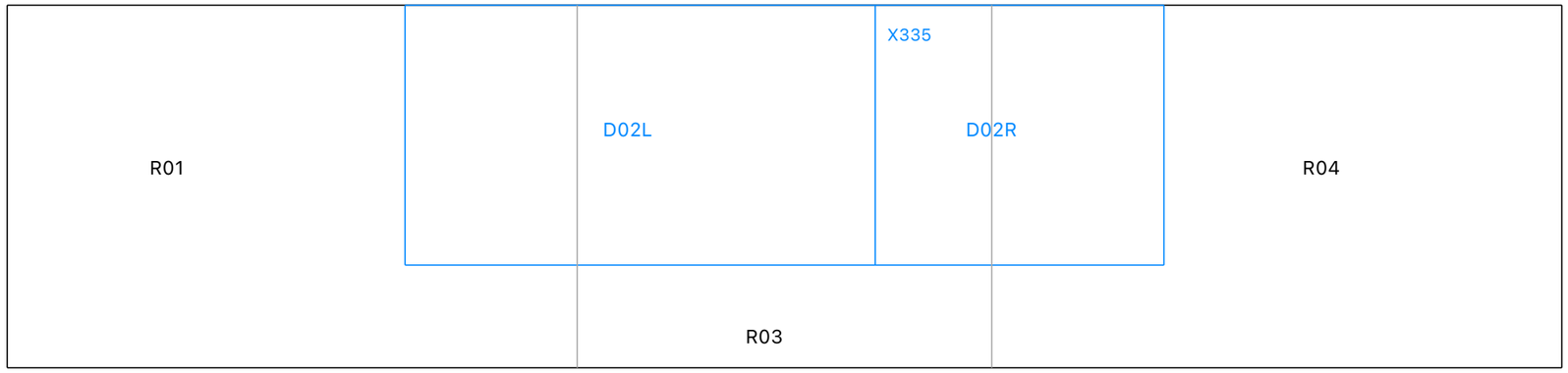
PETG; nominal 250 cube, 0.4 nozzle. Usable margins, layer, support, tolerance and calibration TBD.

No strength, physical fit, complete drawing/model approval or fabrication release is implied.

Sources: B24-JOINTS-02 definition; B26-DETAIL-01; B26-ROOF-01; B26-PANELS-02 pp5-6.

1 / Roof boundaries and panel mating

PLAN: nominal interfaces before apertures; roof D02L/D02R in blue



D02L outer X153.55..335; D02R X335..446.45. Both occupy Y0..100.346434.

R01 ends at X153.55; R04 begins at X446.45 for Y0..100.346434; R03 begins

at Y100.346434 for X220..380. R01/R04 keep the B26-PANELS-02 stepped jog seams.

The roof and panels remain separate parts; no new roof-owned shelf under R01/R04.

The roof's X153.55/446.45 side edges meet the panel edges, not a fused panel joint.

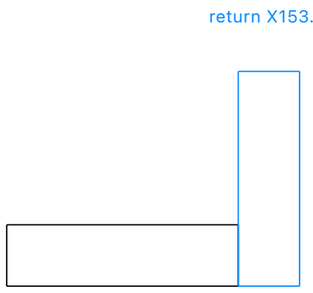
X220/380 are lower-shell seams; they do not split D02L/D02R or alter their X335 seam.

Accepted roof geometry controls where it conflicts with the unapproved panel draft.

Scope excludes the screen opening, toes, connectors, controls and their full solid envelopes.

2 / Side returns and front flange

LEFT TRANSVERSE SECTION / at 2 < Y < 100.346; right is mirrored about X300



D02L return X153.55..155.55, D02R return X444.45..446.45; no new side shelves.

Roof return upper profile is U(Y), lower Z18, Y2..100.346434 as ROOF-01 sheet 4.

For Y2..43.515278, U=85.376766; then U=25+tan(49deg)*(96-Y).

Return touches panel at X153.55 or X446.45, Z18..20; zero nominal edge gap.

R01/R04 retain their shell-owned panel posts. Edge stiffness and physical fit are unverified.

No underside roof shelf is added: it would obstruct vertical roof installation at the panels.

The intended roof-first / R03-last order avoids lowering the front flange through R03.

All roof-to-panel and shell clearances, including assembly path, remain to be tested.

FRONT / two roof-owned flange wedges and two shell-owned posts

At X223..233 and X367..377 extrude Y/Z polygon (95,16), (116,16), (116,18),

(100.346434,18), (95,24.150368), close; upper edge follows roof underside U-2.

Each flange supports R03 underside Z18 over Y100.346434..116; panel is not fused to it.

Axes (228,110)/(372,110): shell D8 post Z2..16; D9-to-D8 foot Z2..5.8, pilot D2.5 Z6..16.

Panel + flange D3.4; panel D6/90deg countersink depth 1.3; nominal BN616 M3x8, top Z20.

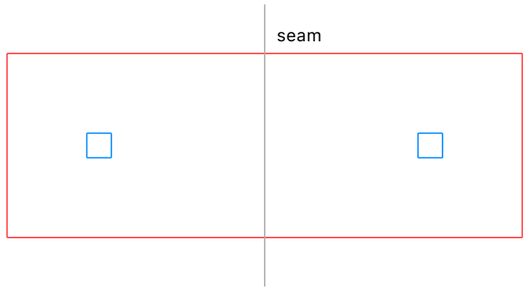
Nominal screw tip Z12; engagement 4 and pilot-bottom reserve 6. No thread or grip qualification.

Place roof before R03 so that the integral front wedges are not lowered through the panel.

Full installation order, display, controls, roof returns and physical access remain untested.

3 / Rear strip, wall and additional seam transfers

REAR STRIP PLAN / each C07 across X220 or X380; drawing not to scale



Add two source-type C07 plates at (220,14) and (380,14), local u along X.

Their XY bounds are X206..234 / 366..394 and Y9..19; Z8..11.

Four integral shell pads: (211,14) C01, (229,14) C02, (371,14) C02, (389,14) C03.

Per pad D9 Z2..8, D2.5 pilot Z2.5..8; plate holes D3.4 at pad axes.

Four additional source-type nominal ISO4762 M3x8 screws; shaft Z3..11, head D5.5 Z11..14.

Calculated engagement 5, tip-bottom reserve 0.5. Total C07 plates 16 and screws 32;

this adds two plates/four pads/four screws to accepted 14-plate B26-JOINTS-03 study.

Each plate and both pads lie inside Y0..20 strip. Wall Y0..2 is not bridged separately.

Rear post foot / new pad minimum plan gap: 1.050; roof post core / plate: 1.000 mm.

Pad-to-wall gap is 9.5-2 = 7.5 mm. These are selected XY bounds, not installed clearance.

LONGITUDINAL SECTION / rear wall and floor are integral in each rear shell

C01/C02/C03 floors extend to Y0, Z0..2; wall Y0..2 is integral to its own floor.

At X220 and X380 the floor and wall meet on butt planes; no continuous fused material crosses

either plane. New C07s connect the floor across each seam; no claimed sealed wall or wall stiffness.

Wall top Z18 outside X153.55..446.45; inside that interval wall top Z83.376766.

Raised wall belongs to C01: X153.55..220; C02: X220..380; C03: X380..446.45.

Its lower and upper wall portions are integral within each owner; openings follow ROOF-01 sheet 4.

Five tall rear posts Y4 X210/230/300/370/390 belong C01/C02/C02/C02/C03 respectively;

D8 core Z2..83.376766, D9-to-D8 foot Z2..5.8, D2.5 pilot Z73.376766..83.376766.

Four outer rear panel posts remain (8,8)/(110,8)/(490,8)/(592,8), per REAR-POST-03.

D02L lap X325..345 Y13..29 Z75.376766..83.376766; D02R holes at (340,18)/(340,24).

Lap holes D3.4, countersink D6/90deg depth 1.3; D02L D2.5 pilot Z76.376766..83.376766.

4 / Sources, precedence and actual unresolved constraints

Authority: docs/APPROVALS.md, shell dimensional delegation and installed-height inputs;

engineer acceptance of B26-JOINTS-03 and F13-STATIC-05 studies; approval of REAR-POST-03.

1. B26-ROOF-01 drawing source sheets 1-4: profile, return, flange, rear posts and lap.

2. B26-REAR-POST-03 source sheet 2 supersedes the old four outer rear post Y positions.

3. B26-JOINTS-03 source sheets 1-3: shell boundaries, 14 transferred C07 joints and section.

4. B26-PANELS-02 source sheets 1-3: unapproved six panel boundary and head layouts.

5. models/enclosure/B24/BUILD123D/panels.py roof() and shell.py post construction;

   B24-JOINTS-02 definition.json bridge/pads/screws: nominal historical source, not measured parts.

New under delegation: the two extra rear-strip C07 joints only; acceptance is not implied.

Higher-precedence named accepted amendments override older draft labels/values.

Fixed platter D110 at (110,165)/(490,165), F13-STATIC-04 foot local Z3.8 vs cover Z4;

no jog/service or purchased part geometry is altered by this interface study.

Remaining blockers: return/control/opening and wall corner full solid intersections;

rear AC inlet, ventilation, equipment and actual connector envelopes against new joints;

roof return/panel approach and access; front flange/display lands and post/wall attachment;

roof lap and post heads, real hardware, seam edge finishing, tools and complete assembly.

Off-case jog service fixture/support and complete installation sweep remain unverified.

Roof Y/Z section and plan checks do not establish 3D collision, physical fit, strength,

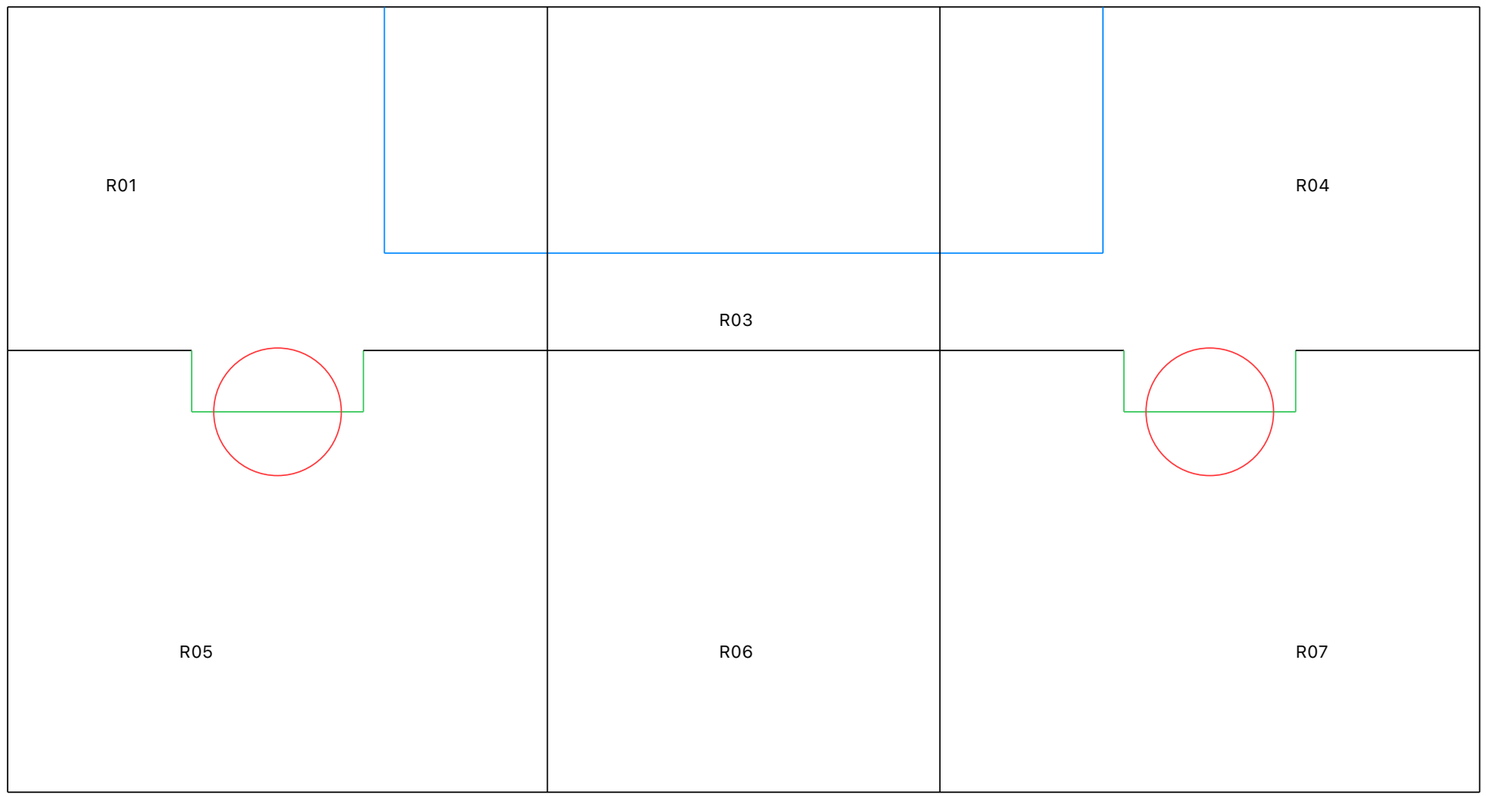
screen retention, water sealing, electrical or thermal safety. No measured cable clearance claimed.

PETG intent, nominal 250 x 250 x 250 bed, 0.4 nozzle; usable margins, orientation, layer,

support, shrinkage, tolerances, thread fit and calibration are unknown. Bounding sizes alone

do not prove printability. No complete named drawing approval or fabrication export is implied.

1 / Six-panel partition with accepted jog seam



Green seam Y165 crosses both jog centres over X75..145 and X455..525, per accepted F13-PANEL-06.

Blue: roof envelope, separate D02L/D02R; red: the two D52 jog passages, divided at their diameters.

Samples: 192000 one-mm plan cells assigned; 100800 D6 head-rim samples remain on their owning panels.

Drawing is a candidate integrated panel definition; it is not an approved complete enclosure drawing.

2 / Exact nominal panel regions and cut precedence

All panel bodies nominal Z18..20, before cuts. Domains are XY from the rear-left case datum.

R01 = [0,220]x[0,140] UNION [75,145]x[140,165], LESS [153.55,220]x[0,100.346434].

R03 = [220,380]x[100.346434,140].

R04 = [380,600]x[0,140] UNION [455,525]x[140,165], LESS [380,446.45]x[0,100.346434].

R05 = [0,220]x[140,320] LESS [75,145]x[140,165]. R06 = [220,380]x[140,320].

R07 = [380,600]x[140,320] LESS [455,525]x[140,165]. Shared boundaries are nominal only.

Cut D52 through each jog panel at (110,165)/(490,165); D108 well cut from Z19 through top.

Use B24 panels.py / B19 fader wells, control profiles and accepted B26 coordinate exceptions.

Transfer old control Y+20; E01/T01/T02, PFL, B01..B10 and P01..P03 use B26-DETAIL-01 exceptions.

Three master pots: X559, old Y21/49/77 +20; B23/B24 centres Y255; E01 (300,125),

T01 (277,125), T02 (323,125). All 49 control cuts use source sizes and actual owner panel.

Preserve old outer-corner/edge treatment where physically applicable; new seam step corners are sharp.

No new seam gap, chamfer, clearance or tolerance is invented; calibrate actual mating separately.

B26-PANELS-02 sheets 1-2 Y145 partition is superseded by accepted F13-PANEL-06 Y165 proposal.

B26-ROOF-01 retains roof boundary and front posts; B26-REAR-POST-03 retains four Y8 rear axes.

Reference only: source outlines are not 3D solid-connectivity, collision, tool-path or physical-fit proof.

3 / Control and closure ownership

Each source control centre and four corners of its conservative aperture rectangle were checked

against the proposed owner. This detects seam/roof crossing only, not body or edge clearance.

Retain the exact cited source profile: conservative triangle rectangles are not the cut shape.

R01: B01, B02, B03, B04, B05

R03: E01, T01, T02

R04: B06, B07, B08, B09, B10, P01, P02, P03

R05: B11, B12, B13, B14, B15, B16, F01

R06: B23, B24, F03, F04, F05, P04, P05, P06, P07, P08, P09, P10, P11, P12, P13, T03, T04, T05, T06

R07: B17, B18, B19, B20, B21, B22, F02

28 CLOSURE AXES / existing source-type flat panel fasteners

R01 (8,8),(110,8),(8,132),(160,132),(212,132). R03 (228,110),(372,110).

R04 (490,8),(592,8),(388,132),(440,132),(592,132).

R05 (8,148),(212,148),(8,230),(212,230),(8,312),(212,312).

R06 (228,148),(372,148),(228,312),(372,312).

R07 (388,148),(592,148),(388,230),(592,230),(388,312),(592,312).

At each axis: panel D3.4 through, D6 90deg countersink from Z20, nominal depth1.3; BN616 M3x8.

Roof has five rear post holes and two lap holes separately; R03 front posts end Z16, remaining Z18.

Source pilot D2.5, source-type D8 core/D9 foot. Installed hardware and print threads require trials.

4 / Parts, source mapping and unresolved checks

Printed proposed controller: six distinct lower shells C01..C06, six distinct flat panels, two roof halves.

C07 bridges x16 / nominal M3x8 x32: accepted B26-JOINTS-03 x14 plus B26-INTEGRATION-01 x2.

Other proposed B24 counts: ten fader end bridges, 30 switch retainers, 13 pot retainers, one encoder retainer, four display toes and 49 caps; add two static F13 jogs and four sensor toes. Counts not yet verified against evaluated geometry or printable exports. U1-U4 board mounting remains deferred.

Printed/purchased/reference roles and remaining screw quantities follow cited drawings and source BOM.

Primary inputs: accepted B26-DETAIL-01, B26-ROOF-01, B26-REAR-POST-03, B26-JOINTS-03, B26-INTEGRATION-01, F13-STATIC-02/04 and F13-PANEL-06; B26-PANELS-02 prior working data.

Exact retained geometry: models/enclosure/B24/BUILD123D/{shell,panels,controls,caps,bridges}.py; B09 controls.json, B19 definition, B22 definition and validation; R12 coordinates CSV and B24 moves.

Source snapshot hashes and dimensional authority are recorded in docs/APPROVALS.md.

Remaining: post/solid connectivity, complete roof and neighboring control interference, shaft/collar tools, full jog and panel installation/removal path, realistic bought bodies/heads, service and fit qualification.

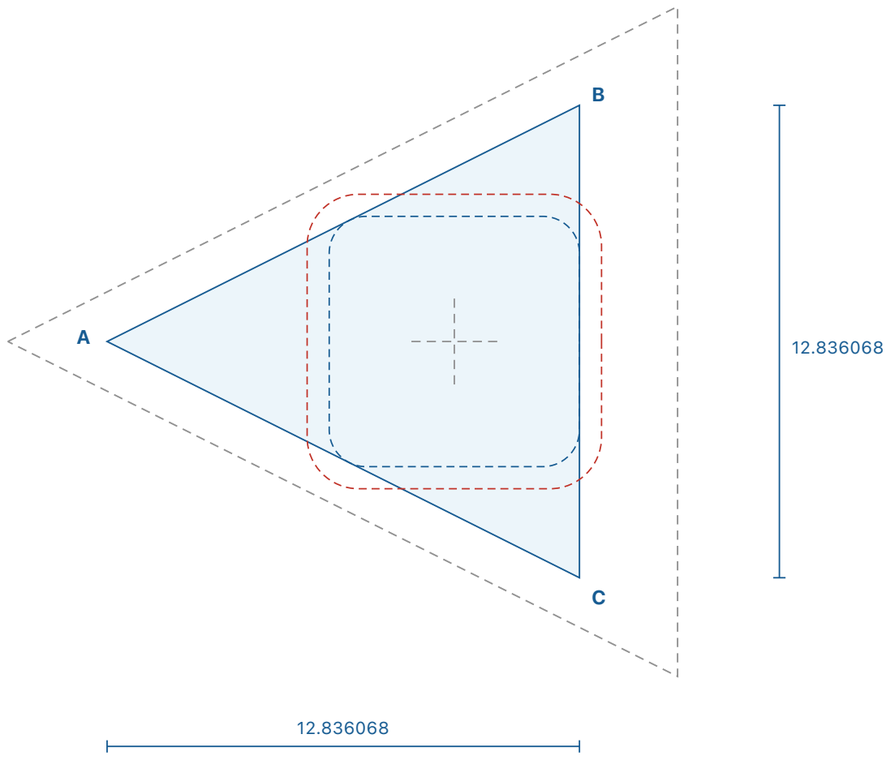
PCB mount holes remain intentionally deferred; cable/plug geometry is engineer-owned, not proven fit.

PETG intent; nominal machine 250 x 250 x 250, nozzle0.4. Usable bed, orientation, supports, layer and calibration are not established. No 3D BRep, visual model, physical fit/strength or export is implied.

01 Minimum triangular head / unchanged mounting interface

Change only the top prism. Preserve the complete source stem/socket below w=0, button axes, occurrence transforms, installed height and panel. C05 is unchanged.

TOP / looking -w / 6:1



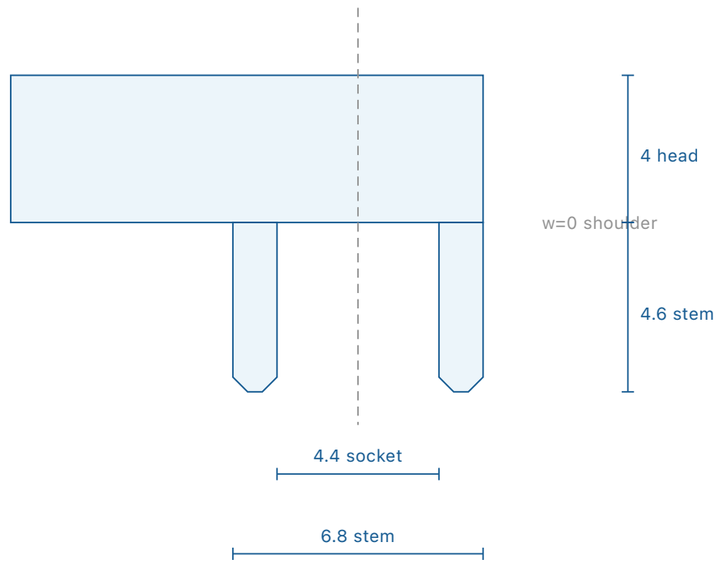
GREY: old18.2 outline / BLUE DASH: stem6.8 R1

RED DASH: unchanged8 R1.4 panel aperture

A(-9.436068,0) / B(3.4,6.418034) / C(3.4,-6.418034). Datum at stem axis, not triangle centroid.
Side equations: u=3.4; -u+2v=a; -u-2v=a. a=7.2+sqrt(5).

Head centroid u=-0.878689 is a derived shape result, NOT movement of the switch or stem axis.
Existing triangle angles and pointing directions remain unchanged.

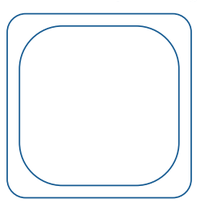
SECTION A-A / v=0 / 6:1



Socket ceiling w=0 / head solid0..4 / total8.6

Both original mouth chamfers remain0.4 high: outer6.8 R1 to6 R0.6; socket4.4 R1 to5.2 R1.4 at w=-4.6. Source faceting/entry geometry is retained exactly, not replaced by these nominal labels.

MOUTH DETAIL / looking +w / 5:1



Exact retained interface: CAP_T_Rev_B16_Fusion01.stl below w=0, plus B16 native F3D. Source hashes in validation JSON. No mating-fit/tolerance change; no inference from drawing scale.

mm / third-angle orthographic / labelled local u,v,w / numerical definitions govern rounded labels / no new tolerance selected

02 Assembly datums / aperture exposure / validation scope

This is a local CAP_T amendment, not a replacement complete enclosure drawing. Read with accepted B09 and delivered B16-Fusion-01. Quantity6 printed; purchased switches/panel are unchanged references.

UNCHANGED INSTALLED POSES / table X,Y,Z

ID	X	Y	shoulderZ	topZ	Fusion rot
T01	277	105	54.25	58.25	0 deg
T02	323	105	54.25	58.25	180 deg
T03	236	258	53.5	57.5	0 deg
T04	254	258	53.5	57.5	180 deg
T05	348	258	53.5	57.5	0 deg
T06	366	258	53.5	57.5	180 deg

Global Fusion position=(tableX,-tableY,shoulderZ). All positions/Z and stem/socket stay fixed. New head is the convex hull A-B-C extruded w=0..4, joined to the exact retained source interface w=-4.6..0. T02/T04/T06 use the existing180-degree occurrence rotation.

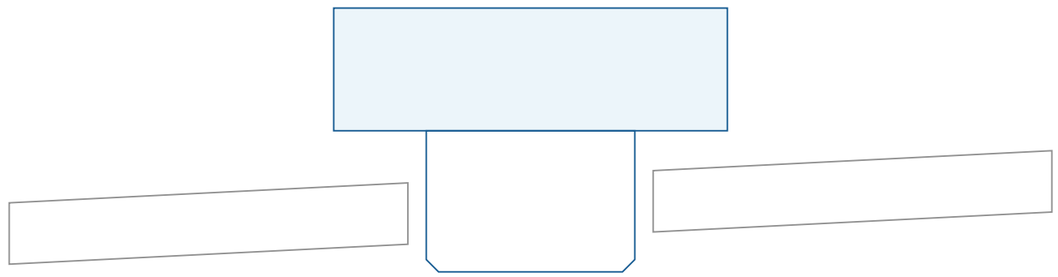
CONSTRUCTION / MINIMALITY

Rounded stem support: c=3.4-1=2.4. Tight sloping supports are a=3c+sqrt(5)=9.4360679775; tight base b=3.4. Preserved side slopes give L=a+b=12.8360679775. All three support lines touch the nominal stem. Smaller parallel offsets cut it. No additional margin is selected.

Source mesh supports agree within1e-6 mm numerical readback (base roundoff below1e-7 mm); this is NOT a manufactured tolerance. Nominal corners are not shrunk to exploit mesh faceting. Head area reduces50.258229%; stem fit and all heights stay unchanged. No motion/physical pass.

BOM: CAP_T printed x6. Cap material TBD; retain prior accessory scope. FDM250 cubed/P2S0.4 nozzle known. Confirm material, layers, usable margins, supports/orientation and calibrated fit before printing. Enclosure PETG does not silently select cap material.

FREE-STATE PANEL CLEARANCE / projected elevation / 5:1



Panel2 vertical / upper slope0.05 at T01/T02

Nominal minimum free gap to the panel plane: T01/T02=1.5-0.05xL/2=1.179098; T03-T06 flat panel=1.5. This is not a certified press stroke. Measured switch travel/return, tolerances and loaded clearance remain TBD.

APERTURE COVER CHANGES — ENGINEER REVIEW

The smaller head no longer covers the entire unchanged8x8 R1.4 opening. Maximum exposed extent normal to sloping edges is0.668328; at the base it is0.6. No hole is reduced, no axis moved, and no new dust/light sealing claim is made. Explicit acceptance of this visible opening is required with drawing approval.

Actual source geometry has been inspected read-only. The live879-entry source is saved/modified at inspection, unlike the earlier unsaved handoff. It is not saved, overwritten or closed here. Existing C05 and all B16 Fusion files remain unchanged. Drawing approval precedes native revision; named model approval precedes STL/STEP/F3D exports.

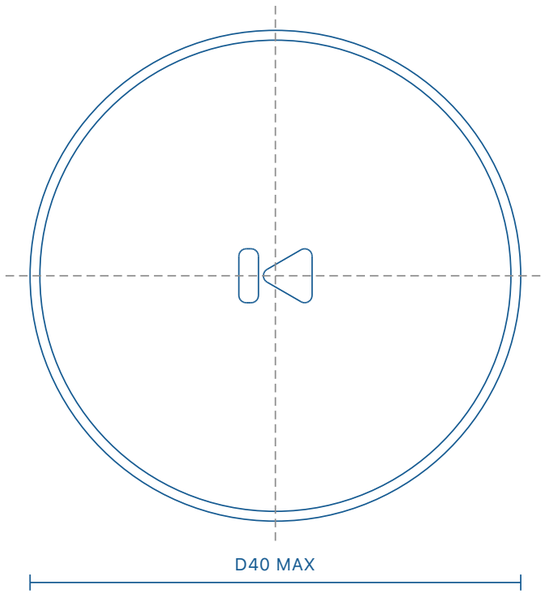
mm / third-angle orthographic / labelled local u,v,w / numerical definitions govern rounded labels / no new tolerance selected

01 Corrected keycap family / source symbols and enlarged HC ring

Supersedes unapproved K01 marking detail only. Keep all head exteriors, heights, stem/socket geometry and B24 positions. CUE/PLAY source logos transfer1:1; HC outline enlargement is explicitly selected by the engineer.

CAP-CUE40 / PRINTED x2

TOP / looking -w / 2:1

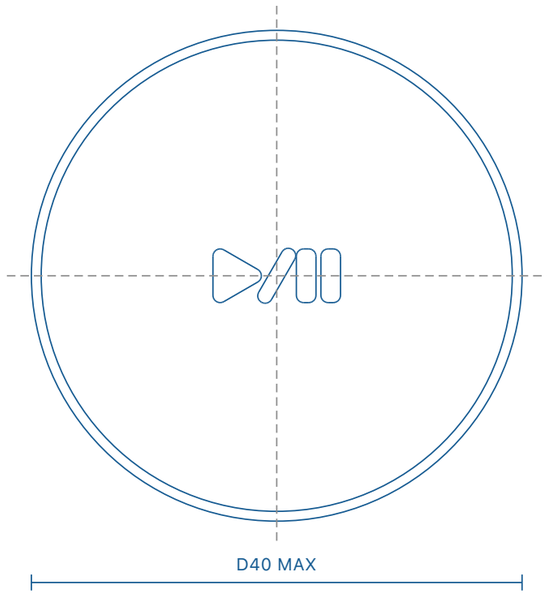


Bottom D38.8 / top D38.4

B11 / B17: source CUE, not text

CAP-PP40 / PRINTED x2

TOP / looking -w / 2:1

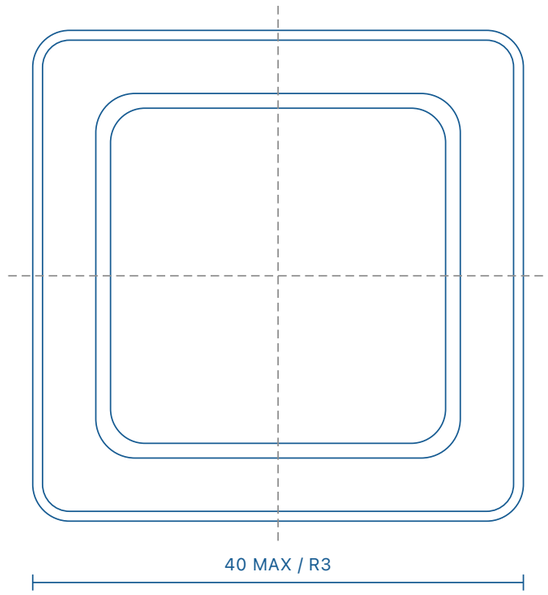


Bottom D38.8 / top D38.4

B12 / B18: source PLAY / slash / PAUSE

CAP-HC40 / PRINTED x8

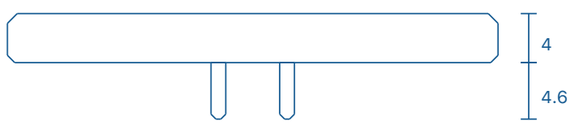
TOP / looking -w / 2:1



Bottom38.8 R2.4 / top38.4 R2.2

B13-B16 / B19-B22: ring29.714286

COMMON HEAD SECTION / v0 / 2:1



Marks omitted here; depth detail sheet2

Headw0..4: lower0.6x45, constant maximum sectionw0.6..3.2, upper0.8x45. HC corner centres(+/-17,+/-17); radii2.4/3/3/2.2 atw0/0.6/3.2/4. Linear section interpolation. Total nominal height8.6. Exact source mechanical interface beloww0 is unchanged.

ONLY MARKINGS CHANGE. Remove/fill the former small square ring on CUE and PLAY/PAUSE; do not leave it behind the new symbols. HC retains a square ring, now29.714286 across the outer mouth. Do not copy the legacy3.6mm stems: current stem length4.6 remains fixed.

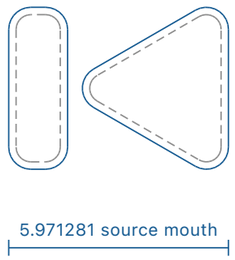
mm / third-angle orthographic / local u,v,w / exact cited source data governs the retained faceted interface and CUE/PLAY voids

NOT APPROVED / NO NEW FUSION GEOMETRY OR STL-STEP-F3D / separate named model approval required

02 Logo definitions / exact source transfer / authorised HC enlargement

Blue: final source mouth atw4. Grey dashed: source floor atw3.6. CUE/PLAY curves come from FreeCAD Sketch003/Pocket001; final STL boundaries include the existing entry/corner details. No fonts, pixel tracing or new symbol design.

CUE / 6:1

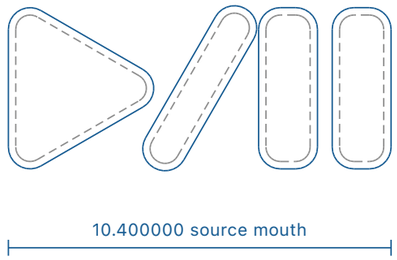


Mouth height 4.400000 / source readback

2 recessed regions: left vertical bar and left-pointing rounded triangle. Keep the source's tiny positional asymmetry aboutu0; no recentering.

Native sketch:14 analytic curves

PLAY / PAUSE / 6:1

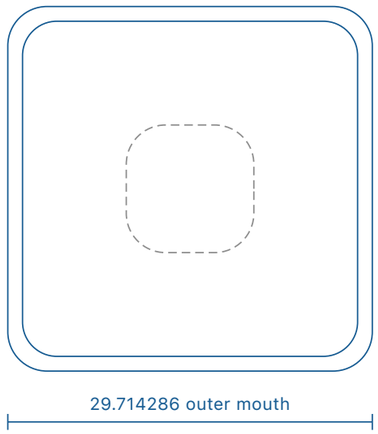


Mouth height 4.481201 / source readback

4 recessed regions: right-pointing rounded triangle, diagonal separator and2 pause bars. Retain all source spacings and corner/entry details.

Native sketch:30 analytic curves

HOT CUE / 2:1



Engineer-selected outline ratio40/14. New outer mouth=10.4x40/14. Straight runs grow; corner radii, groove width0.8, depth0.4 and entry0.2 remain unchanged. Grey: old10.4 outline.

POCKET DEPTH / LOCAL DETAIL / 8:1



w4 top / w3.6 floor / entry0.2x45

CUE/PLAY: final source STL voids transfer1:1 in u/v/w, including facets and actual mouth shape. Native pocket depth0.4 and entry chamfer0.2 are retained; the figure shows axial intent, not a substitute simplified symbol. Minimum nominal solid web abovew0 is3.6.

Source data: DJC_Keycaps_Rev_K02_logo_sources.json contains all native curve coordinates, exact final STL mouth/floor loops, sources and hashes. The source files plus those coordinates uniquely define the symbols. Native/STL vertex agreement is below1e-6mm numerical, not a manufactured tolerance.

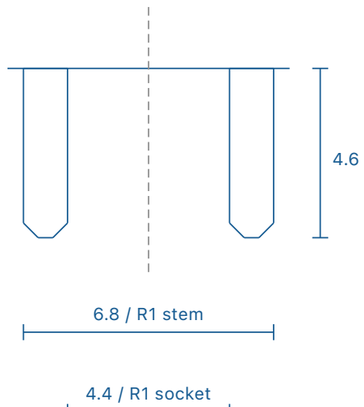
HC sections: w3.6/3.8 outer29.314286 R3, inner27.714286 R3; w4 outer29.714286 R3.2, inner27.314286 R2.8. Delta per side9.657143; all dimensions derive from the source plus selected outline ratio. Straight top rim4.342857=(38.4-29.714286)/2, not strength certification.

mm / third-angle orthographic / local u,v,w / exact cited source data governs the retained faceted interface and CUE/PLAY voids

03 Retained mounting / assembly datums / BOM and checks

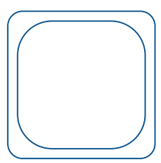
Read with K01's exact-interface definition and the approved B24 assembly. This accessory revision has three distinct printed cap definitions, not a new panel or complete enclosure. No native Fusion model has yet been modified or approved.

STEM SECTION / v0 / 6:1



w-4.2..0: outer6.8 R1 / socket4.4 R1. w-4.6: outer6 R0.6 / mouth5.2 R1.4. Lower lead-ins0.4 axial; socket ceilingw0. Exact current B24/performance-pad faces and lower source details retained; nominal labels do not permit reconstruction or smoothing.

MOUTH / looking +w / 4:1



No change to actual socket, clearance or source faceting.

UNCHANGED POSITIONS / table XY

ID	X	Y	variant
B11	32	228	CUE
B12	32	274	PLAY/PAUSE
B13	78	228	HOT CUE
B14	124	228	HOT CUE
B15	78	274	HOT CUE
B16	124	274	HOT CUE
B17	412	228	CUE
B18	412	274	PLAY/PAUSE
B19	458	228	HOT CUE
B20	504	228	HOT CUE
B21	458	274	HOT CUE
B22	504	274	HOT CUE

Fusion=(X,-Y,21.5+w); head topZ25.5 and stem mouthZ16.9. Flat panelZ18..20; marks can reach20.4. Free gap1.5 to panel / at least1.1 to raised marks. No measured switch stroke or pressed state is invented.

BOM: CAP-CUE40 x2; CAP-PP40 x2; CAP-HC40 x8. Keep original14x14 CAP_B x12 at B01-B10/B23/B24. Do not resize the shared24-instance definition. No bought component, panel opening or mount changes.

DRAWING-STAGE CHECKS

Exact source sketches have closed endpoints.4 PLAY/PAUSE and2 CUE floor loops match native curves within1e-6mm numerical. Source head exterior and3.6mm stem are excluded; only the marking void is transferred. Logo recovery bounds stay inside the old head envelope.

Unchanged new-head boxes:12x397=4764 pairs against all other archived B24 body instances, including32 sheets; no positive AABB overlap above1e-6mm. Nearest new-head box gap6=46-40. This is not native BRep/model collision certification or validation of later live edits.

Material TBD for caps; PLA/PETG available. FDM250 cubed, P2S/nozzle0.4 known. Raw envelope40x40x8.6 fits; usable margins, orientation, layers, supports/brim/raft and calibrated fit remain unqualified. Larger heads require physical eccentric-load/return/retention checks.

Gate: approve drawing B24-KCAP-02, then native standalone B24-KCAP-Fusion-02 and actual Fusion views/checks, then separate model approval before STL/STEP/F3D in models/enclosure/B24/FUSION/KEYCAPS_K02/. Preserve K01 and modified B24; no cloud save or overwrite.

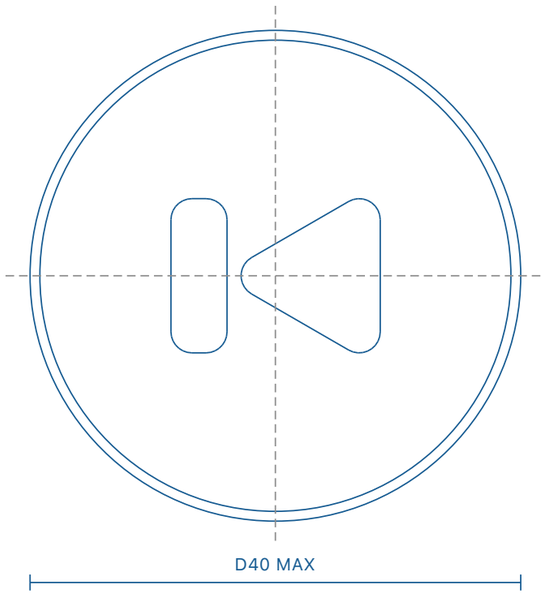
mm / third-angle orthographic / local u,v,w / exact cited source data governs the retained faceted interface and CUE/PLAY voids

01 Larger CUE and PLAY/PAUSE symbols / unchanged cap family

Supersedes only K02's CUE/PLAY logo size. Engineer selects exact20/7 scaling in XY, retaining source shapes and all proportions, depth0.4 and every axial level. HOT CUE, head exteriors, mounting and B24 positions are unchanged.

CAP-CUE40 / PRINTED x2

TOP / looking -w / 2:1

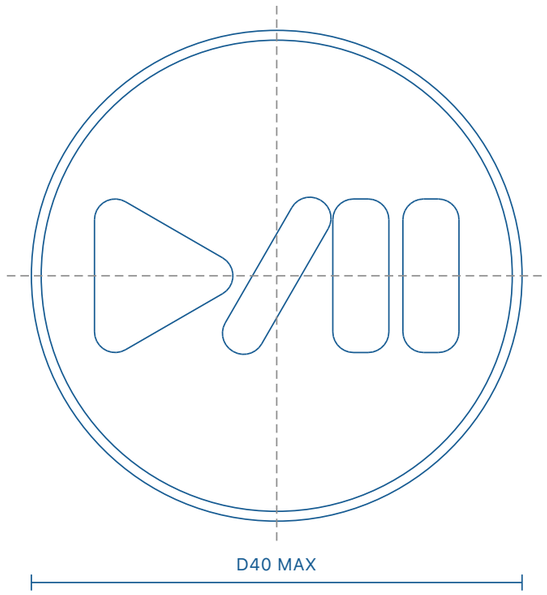


Bottom D38.8 / top D38.4

B11 / B17: source CUE, not text

CAP-PP40 / PRINTED x2

TOP / looking -w / 2:1

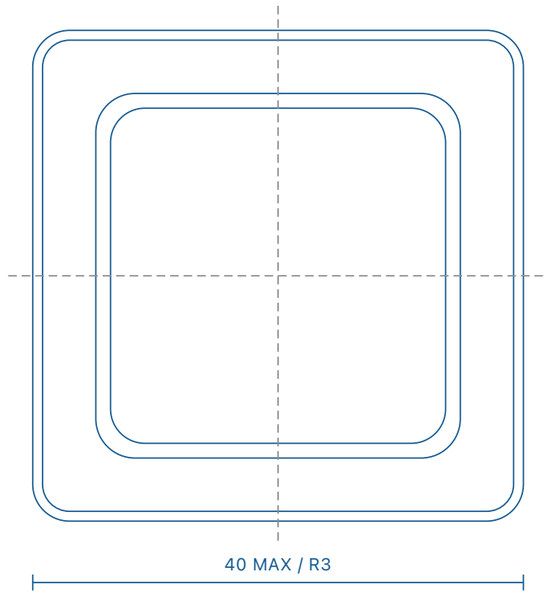


Bottom D38.8 / top D38.4

B12 / B18: source PLAY / slash / PAUSE

CAP-HC40 / PRINTED x8

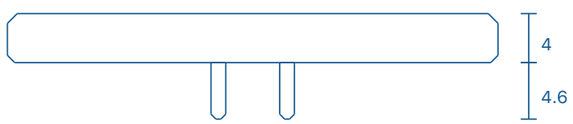
TOP / looking -w / 2:1



Bottom38.8 R2.4 / top38.4 R2.2

B13-B16 / B19-B22: ring29.714286

COMMON HEAD SECTION / v0 / 2:1



Marks omitted here; depth detail sheet2

Headw0..4: lower0.6x45, constant maximum sectionw0.6..3.2, upper0.8x45. HC corner centres(+/-17,+/-17); radii2.4/3/3/2.2 atw0/0.6/3.2/4. Linear section interpolation. Total nominal height8.6. Exact source mechanical interface beloww0 is unchanged.

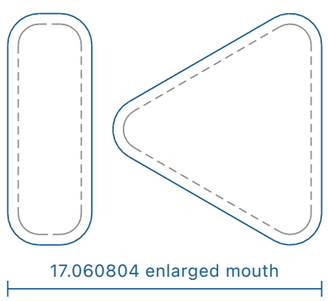
ONLY MARKINGS CHANGE. Remove/fill the former small square ring on CUE and PLAY/PAUSE; do not leave it behind the new symbols. HC retains a square ring, now29.714286 across the outer mouth. Do not copy the legacy3.6mm stems: current stem length4.6 remains fixed.

mm / third-angle orthographic / local u,v,w / CUE-PLAY source voids transformed by20/7 in XY only / exact source mounting retained

02 Enlarged logo definitions / exact source shape and XY transform

Blue: scaled source mouth atw4. Grey dashed: scaled source floor atw3.6. For the entire CUE/PLAY marking void use(u,v,w)->(20u/7,20v/7,w). No font substitution, recentering, mirroring or selective stretching.

CUE / 3:1

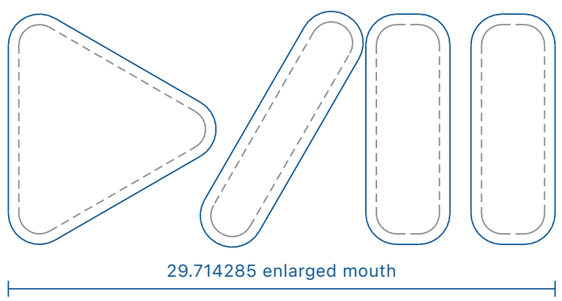


Mouth height 12.571429 / source x20/7

Same2 regions: left bar and left-pointing triangle. All source XY coordinates, radii, spacing and tiny positional asymmetry scale20/7 aboutu0; no recentering.

Source14 curves / XY x20/7 / Z x1

PLAY / PAUSE / 3:1

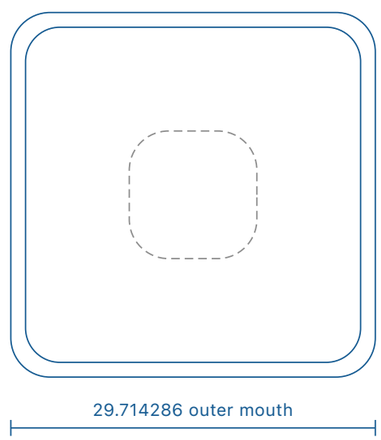


Mouth height 12.803431 / source x20/7

Same4 regions: right-pointing play triangle, diagonal slash and2 pause bars. All source XY coordinates, radii and spacings scale20/7; no shape changes.

Source30 curves / XY x20/7 / Z x1

HOT CUE / 2:1



Engineer-selected outline ratio40/14. New outer mouth=10.4x40/14. Straight runs grow; corner radii, groove width0.8, depth0.4 and entry0.2 remain unchanged. Grey: old10.4 outline.

HOT CUE POCKET SECTION / 8:1



HC only: unchanged entry0.2x45

CUE/PLAY: depth stays0.4, floorw3.6, entry height0.2. Source0.2 nominal lateral entry offset becomes0.571429 under XY scaling. It is NOT0.2x45 after scaling; do not rechamfer. Whole source void faces/facets follow the affine transform. Minimum nominal head web remains3.6.

Unchanged K02 source data contains native curves and final STL mouth/floor loops. Apply20/7 to every sourceu/v value; leavew unchanged. This completely defines the larger symbols while preserving all planar proportions. Displayed envelopes are numerical readback, not fit tolerances.

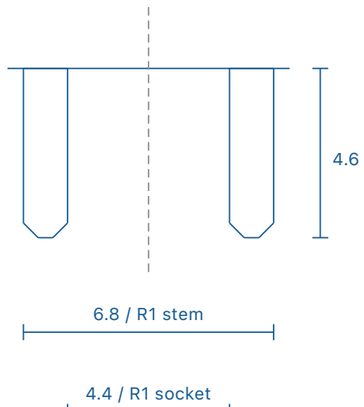
HC sections: w3.6/3.8 outer29.314286 R3, inner27.714286 R3; w4 outer29.714286 R3.2, inner27.314286 R2.8. Delta per side9.657143; all dimensions derive from the source plus selected outline ratio. Straight top rim4.342857=(38.4-29.714286)/2, not strength certification.

mm / third-angle orthographic / local u,v,w / CUE-PLAY source voids transformed by20/7 in XY only / exact source mounting retained

03 Retained mounting / assembly datums / BOM and checks

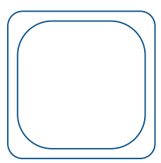
Read with K01's exact-interface definition and the approved B24 assembly. This accessory revision has three distinct printed cap definitions, not a new panel or complete enclosure. No native Fusion model has yet been modified or approved.

STEM SECTION / v0 / 6:1



w-4.2..0: outer6.8 R1 / socket4.4 R1. w-4.6: outer6 R0.6 / mouth5.2 R1.4. Lower lead-ins0.4 axial; socket ceilingw0. Exact current B24/performance-pad faces and lower source details retained; nominal labels do not permit reconstruction or smoothing.

MOUTH / looking +w / 4:1



No change to actual socket, clearance or source faceting.

UNCHANGED POSITIONS / table XY

ID	X	Y	variant
B11	32	228	CUE
B12	32	274	PLAY/PAUSE
B13	78	228	HOT CUE
B14	124	228	HOT CUE
B15	78	274	HOT CUE
B16	124	274	HOT CUE
B17	412	228	CUE
B18	412	274	PLAY/PAUSE
B19	458	228	HOT CUE
B20	504	228	HOT CUE
B21	458	274	HOT CUE
B22	504	274	HOT CUE

Fusion=(X,-Y,21.5+w); head topZ25.5 and stem mouthZ16.9. Flat panelZ18..20; marks can reach20.4. Free gap1.5 to panel / at least1.1 to raised marks. No measured switch stroke or pressed state is invented.

BOM: CAP-CUE40 x2; CAP-PP40 x2; CAP-HC40 x8. Keep original14x14 CAP_B x12 at B01-B10/B23/B24. Do not resize the shared24-instance definition. No bought component, panel opening or mount changes.

DRAWING-STAGE CHECKS

K02 source shapes are retained, not redrawn. CUE/PLAY2D contours scale exactly20/7, Z remains1; both enlarged marking bounds fit within the new circular topR19.2. Source recovery occurs before scaling; legacy head/stem geometry is excluded. No new native Fusion solid or physical pass.

Unchanged new-head boxes:12x397=4764 pairs against all other archived B24 body instances, including32 sheets; no positive AABB overlap above1e-6mm. Nearest new-head box gap6=46-40. This is not native BRep/model collision certification or validation of later live edits.

Material TBD for caps; PLA/PETG available. FDM250 cubed, P2S/nozzle0.4 known. Raw envelope40x40x8.6 fits; usable margins, orientation, layers, supports/brim/raft and calibrated fit remain unqualified. Larger heads require physical eccentric-load/return/retention checks.

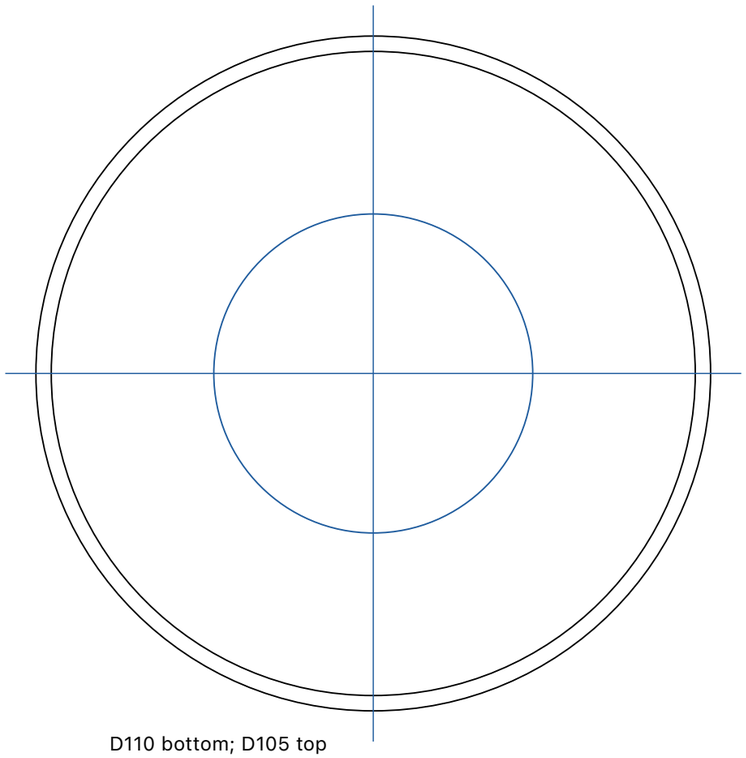
Gate: approve drawing B24-KCAP-03, then native standalone B24-KCAP-Fusion-03 and actual Fusion views/checks, then separate model approval before STL/STEP/F3D in models/enclosure/B24/FUSION/KEYCAPS_K03/. Preserve K01/K02 and modified B24; no cloud save or overwrite.

mm / third-angle orthographic / local u,v,w / CUE-PLAY source voids transformed by20/7 in XY only / exact source mounting retained

NOT APPROVED / NO NEW FUSION GEOMETRY OR STL-STEP-F3D / separate named model approval required

01 Static jog arrangement and hard integration blockers

PLAN / schematic / 1:1



D110 bottom; D105 top

D52 inherited panel aperture REF

AXIAL DATUM / values govern; outlines schematic

Local O = axis on inside floor. +Z up; local angles CCW from +X.

Floor -2..0, panel 16..18 are inherited references, not B26 confirmation.

PLATTER 21.4..36.4, lower D110, upper D105; thickness 15.

Bearings 5.5..12.5 and 12.5..19.5; SKF 61806-2RS1 D30/D42 x 7.

Shaft D29.95, bore D26; carrier bore D42.15: nominal radial gaps

0.025 and 0.075 respectively. Actual corner/seat geometry is unverified.

Outer races bounded by covers Z4..5.5 and Z19.5..21; D40 openings.

Inner races bounded by shoulder Z19.5 and rotating collar at Z5.5.

Magnet face Z6.6; nominal IC face Z4.1: gap 2.5, not a test result.

No axial travel, return springs, pusher, switch, or switch mounting.

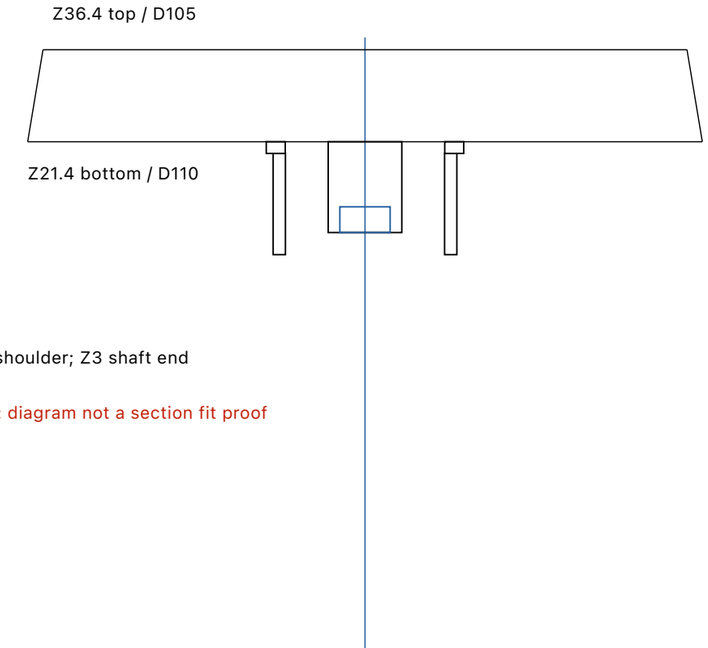
The bearing carrier must be fixed, not a pressable moving cassette.

CRITICAL OPEN INTERFACES — DO NOT MODEL AS A COMPLETE ASSEMBLY

1. Independent sensor support and attachment to carrier/floor: geometry, hardware, installation and wiring TBD.
2. Carrier-to-floor installation: floor screws at R31 lie under the D110 platter; a usable tool/assembly path is TBD.
3. Captured panel sequence and screw-driver paths are not proven. The assembly shown is not yet constructible.
4. D110 bottom exceeds inherited D108 x 1 well diameter; its clearance and intended appearance need review.
5. PCB laminate 2 mm is engineer-confirmed; components, header, wire bend, bearing corners, print margins and fit are TBD.
6. Floor, panel, roof, seams and neighbor control clearances need a complete integrated drawing.

02 Rotor, bearing races and nominal axial chain

ROTOR SIDE / 1:1, local Z / symbolic section



NOTE: diagram not a section fit proof

P01, 1/jog, rotating: solid frustum axis Z, bottom D110 at Z21.4,
top D105 at Z36.4. No overall scaling of spindle or magnet pocket.
Hollow shaft: D29.95 outside, D26 inside, Z3..21.4; union to platter.
Top bearing shoulder: D32.2 outside/D26 inside, Z19.5..21.4;
root edge R0.2 maximum, not across bearing or seal contact.
Centre post D12, Z6.6..21.4, joins platter. Blind socket D8.2,
Z6.6..10.8, opens down. Magnet D8 x 4, face flush Z6.6, top Z10.6;
0.2 socket ceiling gap. Adhesive and magnet retention need qualification.
T30 male: Z3..5.5; RH, pitch 0.5, crest R14.975, root R14.725.
Let t = (Z - 0.5*theta/360) wrapped into [-0.25,0.25).
R=14.975 for |t|<=0.03125; descend at tan(60 deg) to R14.725
at |t|=0.1755875673; root thereafter. Not ISO thread compliance.
P04/P07 female radius = male radius +0.05 at same global Z phase.
P04 D32.2 Z4..5.5; P07 D32.2 Z3..4. 3 grip slots each:
R15.6..16.2, tangential +/-1, angles 0/120/240, full height.
Set P04 to first contact, jam P07 while holding P04. No preload.
Rotor motion is rotational only; endplay and loosening require trials.

DECORATION: fully specified adaptations from F12-D05 exterior

12 radial D10 cylinder recesses at 0+30*k deg, k=0..11, axis horizontal, centre Z28.9; tool radial extent 50.25..60.

At each inner start edge use R1 fillet, leaving D8 flat. This translates each source side tool outward 5 and up 1.5.

Top C groove: bottom Z35.6, annulus R16.8..19.2 minus X[-2,2] only for Y>0; fillet four end corners R0.8.

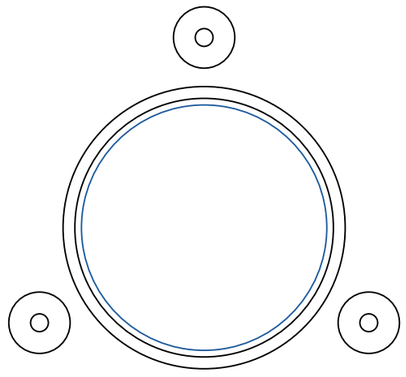
Opening Z36.4 is a 0.8 outward offset of the complete bottom profile, including ends; ruled loft and cut above top.

Only exterior details move: shaft, thread, magnet and bearing interfaces remain at the predecessor datums.

All uncalled edges sharp. Section/printability/topology of these proposed cuts is not verified.

03 Fixed carrier, covers, mounting and sensor interfaces

FIXED CARRIER / plan / schematic



D46 / bore D42.15; ears R31

Ears and bridges are proposals; connections not checked.

P02, 1/jog, FIXED candidate: D46 cylinder Z5.5..19.5, through bore D42.15.

Cover bosses D5 at R23, angles 0/120/240: Z5.5..9.5 and Z15.5..19.5.

Union bosses, then recut D42.15 bore. D1.6 pilots on same axes:

lower Z5.5..8.5; upper Z16.5..19.5. Trial M2 screw forming only.

Three ear caps D10, axis R31 at 90/210/330 deg, Z6..9; through D2.9.

Each radial bridge r22..31, tangent +/-5, Z6..9, united to carrier;

recut bearing bore after union. Base or spindle contact is not assumed.

Floor boss REF D10 x 6, trial pilot D2 x 5, 3 x M2.5 x 6 screw:

ear thickness 3 -> nominal engagement 3 and blind reserve 2.

Head D4.5 max, height 2.5 max, hex key 2; seat min is not supplied.

P05/P06, 1 each fixed: D46/D40 annulus; Z19.5..21 / Z4..5.5.

Add three D5 caps at R23/0,120,240; recut D40. D2.4 through;

D4.2 90-deg countersink depth 0.9 from outer face, 3 per cover.

6 x Newstar M2x4 DIN7991 D4/K1.2 nominal. No spring reliefs.

Faces bear outer race only; D40 opening and actual seals must be checked.

Carrier is fixed to floor, not suspended from a panel. Mounting and

assembly are BLOCKED until sensor support and tool paths are resolved.

SENSOR — existing module is retained, support is NOT DEFINED

AS5048A module REF 15 x 17, R9 contour, 2 x D2.2 holes on 12 pitch; hole axes (+/-6,0), U=+X, V=-Y.

Source: https://www.fruugo.pl/2x-as5048a-enkoder-magnetyczny-interfejs-pwmspi-wysokiej-precyzji-14-bitowy-silnik-bezszczotkowy-enkoder-as5048a/p-532477684-1079695071

PCB back Z1/top Z3 uses engineer-confirmed laminate 2; IC projection 1.1 is nominal, not measured.

Prior toe contact and 2 x M2 x 4 screws / metal M2 nuts are references, not a complete support definition for this design.

Opening in shaft D26 gives nominal radius 13; prior toe rectangular corner radius 12.6495, radial margin 0.3505.

Header and solder/wire volume, cable exit, mounting method, sensor-bearing/shaft collision and driver access remain TBD.

04 Parts list, sequence, derivations and limits

PER JOG / TWO IDENTICAL JOGS PER CONTROLLER

Rotating printed PETG intended / PLA trial: P01 platter+shaft+magnet post x1; P04 collar x1; P07 jam collar x1.

Fixed printed PETG intended / PLA trial: P02 carrier with ears x1; P05 upper cover x1; P06 lower cover x1.

Fixed sensor support / fasteners: TBD — prior P08 toes x2 and M2 fasteners are reference only, not selected final BOM.

Purchased: SKF 61806-2RS1 D30 x D42 x 7 bearings x2; diametric D8 x 4 magnet x1; AS5048A module x1.

Purchased: M2x4 DIN7991 countersunk x6; M2.5x6 socket head x3. Floor bosses are enclosure, not extra jog parts.

No spring, microswitch, pusher, H05 retainer or toggle hardware. No mounting or position for deferred toggle.

PROVISIONAL CASSETTE ASSEMBLY — NOT AN INSTALLATION PROCEDURE

A. With carrier out of case, fit bearings; install covers with heads accessible from their respective outside faces.

B. Through detached panel D52 opening, insert rotor into cassette from above; install P04 and P07 from below.
Hold P04 while setting P07. Axial/radial tool envelope for the notches is TBD; no tool is dimensioned.

C. Only after collar work may a sensor assembly be introduced from below; its independent attachment and wiring are TBD.

D. Mounting ears cannot be assumed accessible: the D110 platter covers all three R31 screw axes. The screw heads are
at Z9..11.5 while the platter begins Z21.4. The D52 aperture and closed floor do not provide a demonstrated path.

E. To disassemble: unresolved until a real mount/tool path exists. Do not imply that sensor, collars or screws are serviceable.

At each ear the vertical distance from head top to platter bottom is 21.4-11.5=9.9; this is NOT a driver access proof.

DERIVATIONS / SCOPE

Lower datum unchanged 21.4; 33.4 + 3 = 36.4; 36.4 - 21.4 = 15. Bottom/top diameters selected 110/105.

B26 JA and JB axes Y165, 5 toward rear relative to B24 with the additional rear extension applied. X values from layout.

Retain accepted +60-deg whole-jog pose as proposed: local mount angles 90/210/330 become global 150/270/30.

Three mount offsets in global table XY: (-26.846788,-15.5), (0,+31), (+26.846788,-15.5); +Y toward operator.

B26 preliminarily reports 8 mm nearest K03 envelope gap in plan; no full 3D collision or tool access result.

Nominal platter top is 18.4 above inherited panel top Z18; inherited 15.5 projection cap is exceeded by 2.9.

The 15 mm thickness and raised top are engineer input; 18.4 mm installed projection remains an explicit design consequence.

No fit tolerance, bearing preload, strength, adhesion, screw engagement, printing, magnetic or electrical performance verified.

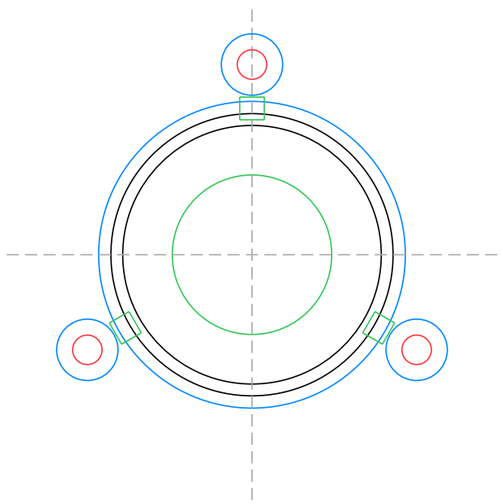
Sources: F12-D05 drawing sheets 01,03,04,08-10,12,14; B26-STUDY-01 sheets 1-2; F06 mounting approval.

Environment metadata only: build123d 0.13.1.dev254+gc49a0493a differs from historic 0.13.0; no Python/CAD used.

Named complete drawing approval, separate model approval, physical validation and manufacturing release are all absent.

1 / Bottom-access jog mounting: plan and leg detail

TOP VIEW / LOCAL XY / 1:1 / maximum plan outlines



Blue: independent sensor plate D50 and floor boss D10

Green: 3 lower feet; red: D4.8 underside head recess

Retain F13-STATIC-01 P01/P04/P07 rotor, P02 bearing bore

and P05/P06 covers unless this amendment explicitly changes them.

P02: keep three R31 ears and their Z6..9 bearing contacts.

Each mounting ear has a D2.0 trial forming bore, local Z6..9.

The screw tip remains 0.5 below the exit; it is a through pilot.

Three feet, each local radial u=22..25.7, tangential v=-2..2,

Z1..4; axes 90/210/330 deg, matching each mounting ear.

Three webs u=24..25.7, v=-2..2, Z4..9; union with feet

and ear bridges u=22..31, v=-5..5, Z6..9, then recut bore.

Feet bear on sensor plate; webs do not occupy lower-cover annulus.

Sensor plate D50 Z0..1 on floor: printed, 1 per jog, removable.

3 floor locator posts D2 at R26.2, 0/120/240 deg, Z0..1.5;

radial plate-to-post nominal gap 0.2. Anti-rotation key on floor:

X24..26/Y-1..1/Z0..1; plate notch X23.8..25.1/Y-1.2..1.2.

Changes to enclosure floor and mounting screws are PROPOSALS, not prior approved geometry.

Maintain local O on the inside floor; +Z up. Floor -2..0, upper panel 16..18 are inherited.

B26 candidate axes table (110,165)/(490,165); old whole-jog +60 deg pose is retained provisionally.

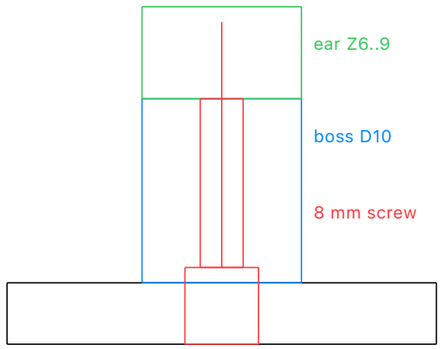
Ear offsets table XY: (-26.846788,-15.5), (0,+31), (+26.846788,-15.5).

Enclosure must provide an independent, correctly registered D10 x 6 boss at each ear.

No toggle button, hole, switch or push stroke is introduced.

2 / Underside fastener and independent sensor plate

SECTION / local Z, schematic; numerical values govern



At each R31 axis: retain an enclosure boss D10, Z0..6,

joined to the 2 mm floor. Cut coaxial underside counterbore

D4.8 from Z-2 to Z0.5, and clearance D2.9 from Z0.5 to Z6.

F13 P02 ear Z6..9: D2.0 through pilot, trial printed thread.

Proposed ETTINGER 001.16.232 M2.5 x 8 ISO4762 A2:

catalogue head D4.5 / height 2.5 / internal hex 2; not measured.

Nominal head Z-2..0.5; tip Z8.5; ear exit Z9: reserve 0.5.

Nominal pilot engagement 2.5; D4.8-D4.5 = 0.3 diametral.

Tool path from underneath must fit D4.8 and actual floor surroundings.

Nominal head bearing and remaining material require physical tests;

2.5 engagement is not a strength or torque acceptance. Actual fastener

and any substitute need engineer acceptance before manufacture.

Sensor plate / use approved F12-D05 references, do not scale or estimate the PCB:

Make disk D50 Z0..1. Reuse F12-D05 P03 PCB locating pegs, toe bosses, nut bays, nut-entry

slots, screw clearance and wire trough, from definition features PCB locating pegs through

Header wire trough; exclude D50 vertical housing, spring guides, stops, retainer and switch wall.

Cut notch X23.8..25.1/Y-1.2..1.2 through plate Z0..1 for the floor anti-rotation key.

Reuse two P08 toes and their M2x4 screws / M2 nuts as conditional references. Their chosen

0.7 mm PCB edge overlap / 3.5 mm^2 bearing area needs engineer and physical acceptance.

Laminate thickness 2 is confirmed; nominal IC face Z4.1 and magnet face Z6.6 -> gap 2.5.

Toe/head/wiring and IC actual envelopes are not supplied. No connected-cable fit is claimed.

Source screw: https://www.ettinger.de/en/product-datasheet/5d91c2de99aa21f87bc1b8445d117b07/create

3 / Assembly, sourcing and unresolved interfaces

1. Install sensor plate, two locating pegs and toe/nut clamps off the controller as defined
   by F12-D05 P03/P08; position the plate with PCB axis coaxial with the retained jog axis.
   Register disk against 3 floor posts and the anti-rotation key defined on sheet 1.
   Confirm fit and PCB support against real copper/components before use.
2. Off-case, assemble both bearings and outer-race covers on P02, insert rotor, fit P04
   and P07 from below, adjust against inner races without preload; real tool fit is TBD.
3. With upper panel off, place the plate, then lower the assembled rotor/cassette over it.
   PCB/toes must enter the D26 spindle without touching it or the magnet. Axial face gap
   at the selected nominal Z values is 2.5, not a magnetic or physical certification.
4. Three feet bear on the plate; ears meet floor bosses. Insert three M2.5 x 8 screws from
   underside through the 3 proposed coaxial bores into P02 ear pilots. Tool envelope,
   insert alignment and floor-face access are not yet tested. The 3 heads sit nominally flush.
5. Proposed six-panel split: near each jog step the panel seam from Y140 to Y145
   between X75..145 (left) and X455..525 (right), then return it to Y140.
   At Y145 the D52 opening chord is 33.2265, versus the D29.95 spindle.
   Slide the front half toward the rear below the platter before installing the roof.
   Plan-only sample: 0..35 mm in 0.1 steps, 301 sections each, 105350 total;
   minimum half-width reserve 1.638 mm. Other bodies and vertical movement excluded.
6. Disassemble in reverse. Real wire bend, disconnect, connector and hand access are TBD.

BOM / PER JOG

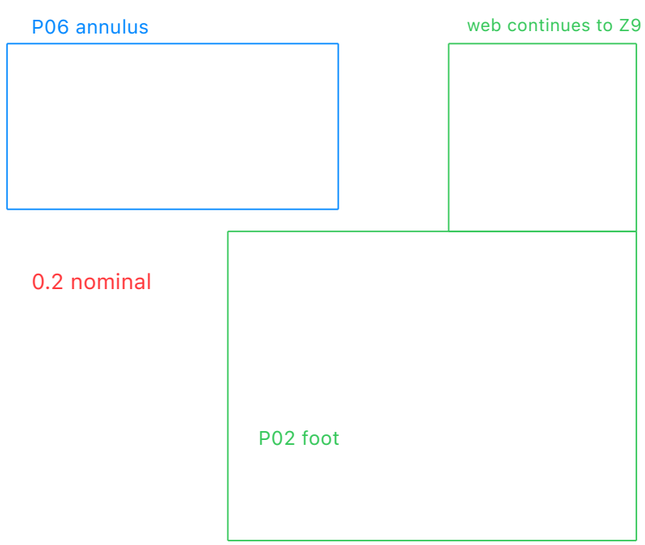
Printed: rotor P01 x1; carrier P02 with feet x1; P04/P07 x1 each; P05/P06 x1 each;
sensor plate x1; F12-D05 P08 toes x2. PETG intended / PLA fit trial, settings TBD.
Purchased: SKF 61806-2RS1 30x42x7 x2; ETTINGER 001.16.232 M2.5x8 x3 PROPOSED;
M2x4 countersunk x6, M2x4 socket x2, M2 nuts x2, AS5048A module x1, D8x4 magnet x1.
Enclosure floor/boss, openings and panel are integral existing-part revisions, not loose BOM parts.

BLOCKERS / NOT READY FOR COMPLETE DRAWING APPROVAL

Actual PCB contact/wire clearance, panel closure sweep around the D110 platter,
underside tool access/strength and complete screen/neighbor interfaces remain open.
No BRep, installed body, strength, print, fit, assembly or magnetic operation was tested.
Plan screen only: 6 D10 boss envelopes vs B25-ELEC-02 4 board rectangles: 24 pairs, minimum gap 24.876 mm.
Same bosses vs 14 B24 C07 bridge boxes: 84 pairs, minimum gap 23.153 mm; other bodies/paths excluded.

1 / Lower foot and cover: clearance amendment

RADIAL SECTION / local foot axis / schematic, dimensions below govern



DATUM: local O is the jog axis on the floor top; case Z = local Z + 2. U radial / V tangential.

For each of the THREE original axes 90, 210, 330 deg: foot U22..25.7, V-2..2, Z1..3.8.

At the same axis: web U24..25.7, V-2..2, Z3.8..9. Join to foot and retained Z6..9 ear bridge.

P06 lower annulus remains D46/D40, Z4..5.5; add its retained D5 cover lobes and cuts unchanged.

Foot top 3.8 to cover underside 4.0 = 0.2 nominal. Web starts outside the annulus at radius24.

Sensor plate D50 Z0..1 and the foot underside Z1 remain unchanged; no bearing or sensor Z moves.

2 / Authority, checks and retained interfaces

AUTHORITY: engineer chose 'Dobierz luz' for the foot/P06 contact. The 0.2 mm nominal prototype

gap is the selected design value, not a measured fit or guaranteed print tolerance. Confirm calibration.

P02: apply the revised foot and web to all three ears. Retain the complete F13-STATIC-02 carrier,

bearing bore, D10 ears R31 Z6..9, ear bridges, D2.0 pilots, underside screws and all other interfaces.

The foot and web share a nominal 1.7 x 4 = 6.8 mm^2 face at Z3.8, before any rounding or tolerance.

At the later ear bridge Z6..9 the web overlaps U24..25.7 / V-2..2; union before re-cutting bearing bore.

P06, both bearings, platter/shaft, covers, collars, sensor plate, magnet and board remain unchanged.

B26 case, panel, jog axis, 8 mm plan HOT CUE gap and no-press mode are not changed by this amendment.

BOUNDED NOMINAL CHECKS

Foot/P06 annulus separation is 0.2 in Z where their plan footprints overlap; web/annulus separation

is at least 1.0 radial, and web/nearest D5 cover-lobe plan separation is 7.840. Other bodies excluded.

Each unchanged foot has 11.893 mm^2 ideal planar overlap with the D50 disk, before notch/fit effects;

positive contact area is not a strength or load qualification. The floor still supports the sensor disk.

Three local interfaces exist per jog, two jogs. Unions, bores, mesh quality and actual faces are not tested.

OPEN / DO NOT MANUFACTURE

0.2 is smaller than many possible FDM errors: prove the gap with a calibration/fit coupon,

and establish acceptance criteria with the engineer. Do not force a cover to seat or add bearing preload.

New F13-STATIC-04 drawing is NOT approved; it does not approve B26, an actual model, STL or STEP.

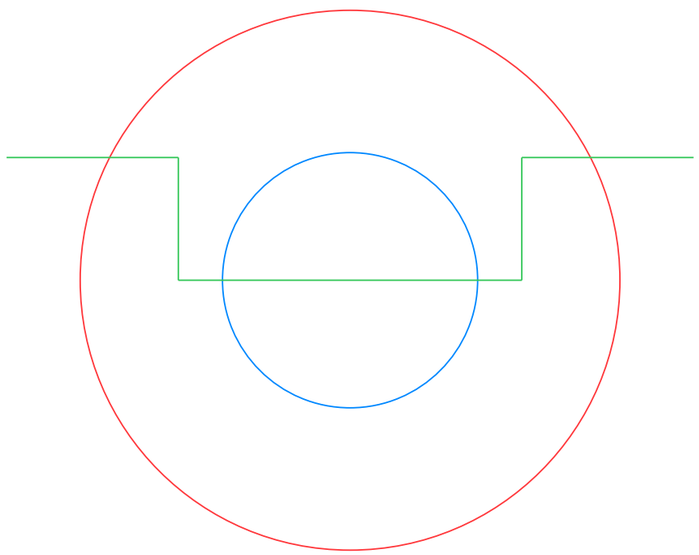
Rear-panel staging, shell/roof integration, tool access, bed margins and full assembly remain unresolved.

No full 3D intersection/distance, motion, physical bearing seating or load test was performed.

Sources: F13-STATIC-02 pp1-3; F13-STATIC-03 p4; F13-STATIC-01 p3; engineer clearance delegation.

1 / Move the seam through the jog axis

PLAN / left jog; right jog is translated +380 X; schematic, dimensions govern



Keep B26 jog centres (110,165) / (490,165), D110 platter and D52 aperture.

Replace the proposed local seam Y145 by Y165 only between X75..145 and X455..525.

Outside those intervals retain Y140; vertical step lines stay at X75/145 and X455/525.

Each jog opening is split through its centre: rear and front panels each have a semicircle.

R01/R04 gain the 70 x 20 strip Y145..165; R05/R07 lose exactly the same strip.

R01/R04 already include the adjacent Y140..145 step from B26-PANELS-02.

All six panels remain distinct. This proposal changes approved F13-STATIC-02 sheet 3:

do not alter its snapshot; this successor would need explicit named drawing approval.

No platter, bearing, floor boss, purchased part, switch or fader axis is changed.

Green: proposed panel seam; blue: D52; red: D110 footprint.

Panel/floor seams, edge finish and calibrated tolerances are not established by this sketch.

2 / Candidate installation without captured rear panel

1. Remove roof and front outer panel; fit the rear outer panel vertically over its control shafts.

   The D52 rear semicircle opens to +Y. Preserve access and retain the panel on its existing posts.

2. Off-case, assemble the complete rotor, bearing carrier, covers and P04/P07 collars before

   sensor installation. Actual simultaneous collar tooling and bearing fit remain unresolved.

3. Hold the jog independently with its centre initially 40 mm forward (+Y) of its final axis.

   Raise it 6 mm in Z from its final pose; shift it rearward 40 mm at that height.

   While moving, the portion of its D51 carrier/cover envelope in the rear panel plane remains

   within the D52 semicircle. Ear caps are below the panel; D110 platter is above the panel.

4. At the final XY axis, lower it 6 mm onto the sensor plate / floor bosses and fasten from below.

   Feet local Z1..3.8 start at Z7..9.8; boss local top Z6: 1 nominal vertical clearance at lift6.

5. With caps removed, raise the front panel 1 mm and approach from +Y, then lower onto its posts.

   D51/D52 plan symmetry provides a bounded necessary check, not a complete swept-solid proof.

   Set the tempo slider as required; verify actual controls, roofs, hardware, wires and tools.

6. Install remaining panels and roof in a tested order. Removal is NOT yet established.

This is a design candidate, not a verified assembly or an instruction to handle an energized device.

The 6 mm lift and 40 mm start are proposed service poses, not permanent geometry or print clearances.

A safe holder for the jog, grip/handling loads, full head and wire envelopes, front controls and roof

paths are not defined. The proposed approach must be checked on the actual complete model and hardware.

3 / Bounded plan calculation and integration limits

Let s be the +Y separation between carrier and final jog axis; 0 <= s <= 40 mm.

Consider only the rear panel material half-plane y <= 0 relative to the final jog axis.

Opening half-width is sqrt(26^2 - y^2); conservative moving D51 circle half-width is

sqrt(25.5^2 - (y-s)^2) wherever both exist. For y<=0 and s>=0, |y-s|>=|y|;

therefore the carrier section is inside the opening and at least 0.5 mm radial reserve exists.

Reflecting about Y165 gives the equivalent front-panel translation check.

Numeric cross-check: 65536 circle-section samples, 0.1 mm shift / 0.05 mm Y; minimum 0.500 mm.

This covers only the maximum circular core/cover-lobe bound at the panel plane, not the ears,

shafts, full Z sweep, chamfers, seams, hardware, adjacent controls, roof, wires or printed variation.

Carrier ears: local Z6..9, final panel underside local Z16. At lift6 ear top Z15: still 1 below.

Lower bearing cover local Z4..5.5; foot and boss heights from F13-STATIC-04 / F13-STATIC-02.

Platter bottom local Z21.4 at final, lifted Z27.4; panel top local Z18. Heights are nominal.

The proposed seam and sequence are unapproved; actual build volume, fixture, floor support,

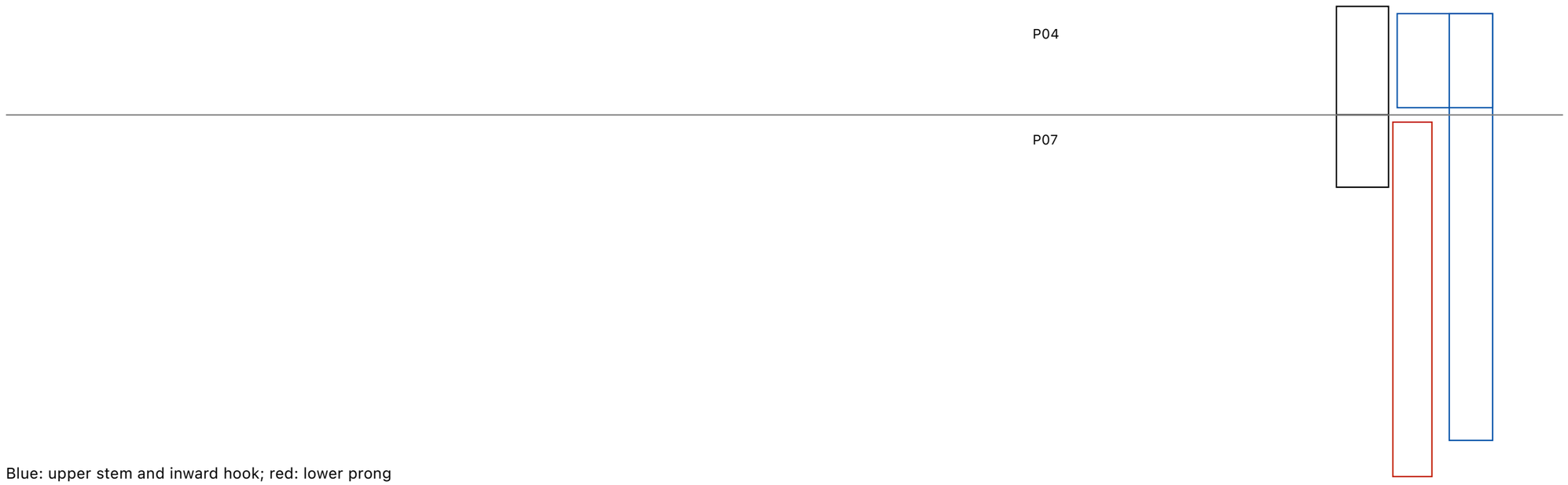
control assembly, screws, body connectivity, fit, strength and complete service remain to be checked.

Sources: F13-STATIC-01/02/04; F13-STATIC-05; B26-PANELS-02; B26-DETAIL-01; B09 controls.

No 3D model, physical fit, electrical safety, strength or final release is implied.

01 Concentric separate upper hold / lower turn interfaces

ENLARGED RADIAL SECTION at one slot; local +Z up; schematic, dimensions govern



Blue: upper stem and inward hook; red: lower prong

Cross-section omits bases, collar thread and cover. See numerical definition below.

NOMINAL TOOL SOLIDS / all coordinates are local to the jog axis

At each angle a=0,120,240 deg: u is outward radial coordinate and v tangential; rotate every box by a.

Upper hold U: three stems u16.3..16.8, v+-0.8, Z-0.5..5.4; inward hooks u15.7..16.8, v+-0.8, Z4.1..5.4.

U base: annulus R16.3..16.9, Z-0.5..0; union all stems. Two opposing handle tabs radial 16.5..25, width 4, same Z.

Lower turn L: three prongs u15.65..16.1, v+-0.8, Z-1..3.9; base annulus R15.6..16.15, Z-1.5..-1.

L base joins prongs; two opposing handle tabs radial 15.9..25, width 4, same Z. Tabs are schematic grasp interfaces only.

All dimensions are selected trial interface envelopes, not tolerances; parts are separate, aligned coaxially. No other solids specified.

02 Installation, analytical limits and outstanding gates

SOURCES AND INTERPRETATION

F13-STATIC-01 sheet 2: P01 D29.95 shaft; T30 RH pitch 0.5; P04 Z4..5.5; P07 Z3..4;

both D32.2; 3 open slots u15.6..16.2, v+-1 through full height. Sheet 1: P06 D40 opening, Z4..5.5.

F13-STATIC-02 sheets 1 and 3 retain these collars/covers; off-case assembly precedes sensor installation.

F12-D05 definition grip-notch coordinates clarify u/v rectangle. Slot phases change with collar rotation.

This study proposes tools only; it changes no approved part and is not an approval of this or any future revision.

IDEAL DIMENSIONAL CHECKS / not a full 3D sweep

U tip inside slot: u15.7..16.2, v+-0.8; minimum root gap 0.10, side gap 0.20; outer part exits open slot.

L prong u15.65..16.1: root gap 0.05, side gap 0.20, slot-end gap 0.10. Penetration inward from R16.1:

U 0.40 at v=0 (at v=+-0.8: 0.380); L 0.45 at v=0 (at v=+-0.8: 0.430); not engagement strength.

U stem to unslotted collar radial gap 16.3-16.1=0.20; to P06 D40 bore: 20-sqrt(16.8^2+0.8^2)=3.1810.

At arbitrary relative angles, U stem to L prong radial gap >= 16.3-sqrt(16.1^2+0.8^2)=0.1801.

At the drawn Z, U hook bottom 4.1 minus L prong top 3.9 = 0.20; U/L bases separate by 0.50 axially.

U hook Z4.1..5.4: 0.10 from each P04 face. L tip Z3.1..3.9 engages P07 0.8 high, 0.10 face margins.

The L prong continues below P07: nominal shaft radial gap >=15.65-14.975=0.675 where shaft exists.

Ideal starting P07 pose: -120 deg and -1/6 mm from the drawn contacting pose, pitch 0.5 RH.

Three 120-deg slots then re-align. P07 Z2.8333..3.8333 overlaps the male thread Z3..5.5 by 0.8333.

Closing +120 deg raises P07 and L 1/6 mm; hook/prong axial gap decreases 0.3667 -> 0.20 mm.

CONDITIONAL OFF-CASE SEQUENCE / IDEAL POSE, NOT PHYSICAL VALIDATION

1. Remove sensor/plate; support cassette off-case. Set P04 to first inner-race contact without preload.

2. Set P07 to the ideal -120 deg / -1/6 mm pose; the three P07 slots align with the P04 slots.

3. Insert U axially through aligned slots; insert L below. Hold U fixed; rotate L +120 deg to contact.

   L rises with P07. Stop at contact; the calculation is not a prescribed tightening torque or force.

4. Withdraw L. For U removal, reverse to the aligned loose pose before withdrawing U axially.

This assumes the F13 same-global-Z ideal thread phase and full helical engagement over the computed path.

Actual thread entry, phase, tool access, stiffness, layer direction, wear, torque and tolerances are unverified.

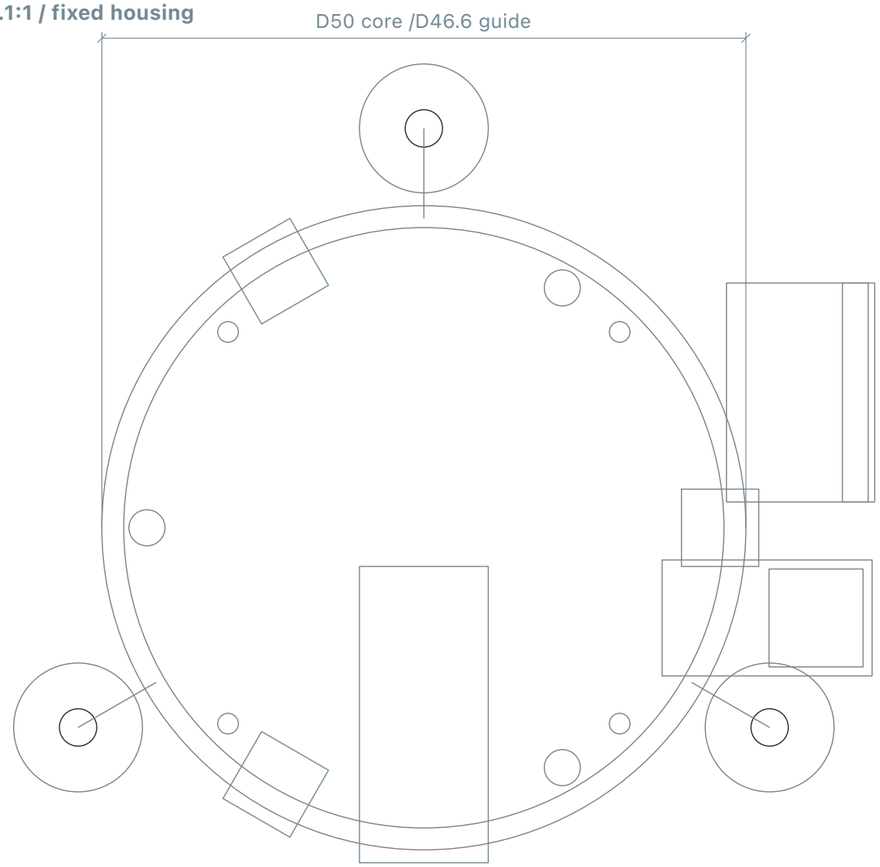
Prong radial widths 0.45/0.5 are near the 0.4 nozzle; printability, strength and real fit require tests.

JOG WHEEL / F12-D05

05 P03 fixed housing: base, ears and guidance

DRAWING NOT APPROVED | DRAFT - NOMINAL PROTOTYPE ONLY - ENGINEER DRAWING APPROVAL REQUIRED

P03 PLAN / 2.1:1 / fixed housing



FEATURE	EXACT NOMINAL DEFINITION
Core / guide / base	D50 Z0..16; boreD46.6 ONLY Z1..16. BaseD50 Z0..1 backs directly on nominal mating support planeZ0.
Ear caps / bridges	3xD10 caps atR31/local90,210,330deg, Z6..9. Bridges radial24..31/tangent+/-5, sameZ. ThroughD2.9 at caps.
Floor-boss clearance	CutD10.4 coaxial with each ear ONLY Z0..6. These admit retained integral floorD10x6 bosses; ear contact stays atZ6.
Three cap/guide channels	At0,120,240deg: cut radial20..26/tangent+/-3/Z2.4..16; open above. Clears P02/P05/P06 D5 caps.
Key channel	Cut X18.6..26.0/Y-10.4..-3.6/Z3.6..16; key has0.4 side/end nominal clearances; lower edge is not press stop.
Upper rail opening	Cut X21.6..26.8/Y-10.4..30.4/Z10.6..16, open above. Main guide retains its other circumferential sectors.
Three structural stops	SolidD2.8 atR21.5/60,180,300deg; Z1..2.5. Entire faces lie under unrecessed P06 annulusR20..23.
Four spring guides	SolidD1.6 atR21.5/45,135,225,315deg; Z1..2.5. Working spring length3.5..2; guide tip clears upper seat0.5 at full press.
PCB locating pegs	TwoD1.8 at(+/-6,0), Z1..2.5; board holesD2.2. Board backZ1/topZ3; source U=+X,V=-Y, header faces-Y.

Three lower stops have full D2.8 contact against P06 lands. Three cap channels and the separate key channel are open from above for straight cassette insertion. Dimensional source choices do not establish stiffness.

Mating surface atZ0 directly backs the1-mm base. Floor bosses remainD10x6 and three M2.5x6 screws. Sheet14 gives the flat B21 coordinate mapping; whole-case neighbors/service remain to be checked.

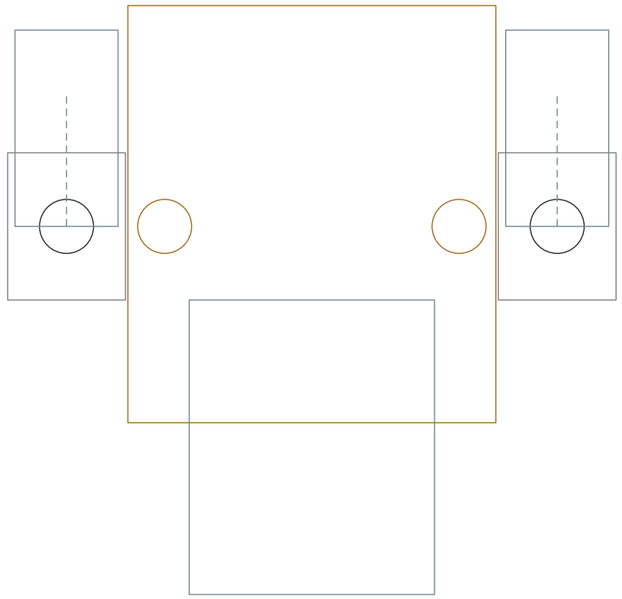
JOG WHEEL / F12-D05

06 P03 sensor support, retainer and cable features

DRAWING NOT APPROVED | DRAFT - NOMINAL PROTOTYPE ONLY - ENGINEER DRAWING APPROVAL REQUIRED

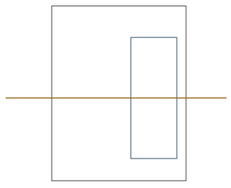
P03 SENSOR / PLAN / 4:1

PCB R9-clipped REF; D04 entries: sheet15



P03 RETAINER / X-Z AT Y-7 / 3:1

Tip24.3 / drive36.3; crestD2.927



FEATURE	EXACT NOMINAL DEFINITION
PCB toe bosses	Two boxes X[7.6,12.4] and[-12.4,-7.6], Y+/-3,Z1..3. Each M2 nut bay AF4.1, Z0.6..2.3, flat faces normalX.
PCB nut entry / bores	Nut bays centred(+/-10,0); slots X relative+/-2.1,Y0..8.0,Z0.6..2.3 open+Y. Blind clearanceD2.2 Z0.2..3 clears M2x4 tip0.3 BELOW nut bay; floor stock0.2. Top load atY5.5 outside bossY3, lower nut bottom3.1->0.65 then slide toY0 without rotation BEFORE PCB/toes/cassette; see sheet15.
Header wire trough	Cut X+/-5/Y-26..-3/Z0..1 through base. Outer exit only Y-26..-23.5 grows toZ3. No header or wire diameter invented.
Retainer base / boss	Base X18.5..34.8/Y-11.5..-2.5/Z0..1; boss X26.8..34.1/Y-10.8..-3.2/Z6..15.5.
Retainer ribs	Boxes X26.8..34.1/Y-10.8..-8.8 andY-5.2..-3.2/Z1..6. Apply floor-boss clearances after unions.
Retainer bore / nut	Axis+X atY-7,Z10.5; boreD3.2 X26.8..34.1. Hex nut pocket AF5.7 X31.1..33.6, flats normalY.
Nut insertion	Open slot X31.1..33.6/Y-9.95..-4.05/Z10.5..15.5; M3 nut AF5.5/H2.4 sits in2.5 axial bay.

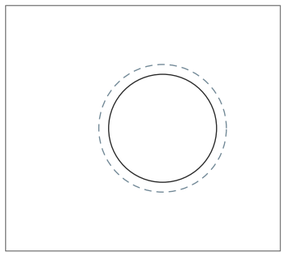
Sensor drawing: supplier15x17/R9 contour; moduleU=+X,V=-Y. Uniform PCB/IC envelopes are references, not a complete component/solder model. Verify actual bare toe-contact areas and1-high wire trough.

JOG WHEEL / F12-D05

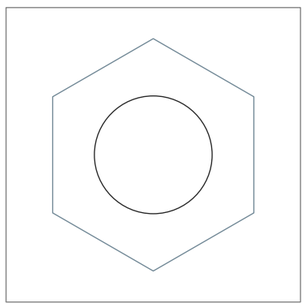
10 P08 sensor toes and P09 captive-nut keepers

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P08 RIGHT TOE / PLAN / 8:1



P09 NUT KEEPER / Y-Z / 8:1



FEATURE	EXACT NOMINAL DEFINITION
Local toe / datum	Right toe: X6.8..12.4/Y-2.5..2.5/Z3..4.3. Install second IDENTICAL part by Rz180 about moduleO. Flat underside3 contacts PCB bare edge.
Hole / head seat	D2.2 through at(+/-10,0); top-entry chamfer0.2x45deg. M2x4 cap headD3.8/H2; toe top4.3, head max6.3.
Contact / backing	Inboard overlap X6.8..7.5 (mirror left) on actual bare board edge; nominal3.5mm2 per toe, backed by base. Verify real components/copper clearance.
Nut / assembly	MetalM2 AF4/H1.6 captive in P03; screw tipZ0.3, nominal projection0.35 below nut, above floor0.3. Do not clamp through an unverified component.
Guide clearance	PCB locating pegs end2.5 below toe3; no extra hole through toe. Maximum rectangular corner radius12.6495 < spindle bore radius13.
Part-local frame	Axis X; datum at nut axis. Box X0..2.4/Y+/-3/Z+/-3. Installed X24.3..26.7; Y7.15/13.65, centreZ4.9 nominal.
Nut pocket	HexAF4.1 X0.7..2.4, regular hex with flat faces normalY; open at+X. Source M2 nutAF4/H1.6. Back web0.7.
Screw clearance	D2.4 through entire part alongX. No chamfers/fillets. Keeps nut from turning; axial screw load seats nut directly against switch body.
Calibration motion	Keeper follows switch Z3.7..6.1. Housing clearance allows full6x6 square. Insert nut from+X before adding switch.

P08 has two mirrored instances; P09 has two identical instances and moves only during switch-height adjustment. Metal nuts remain bought parts, not printed threads. No unsupported screw-head-strength assumption.

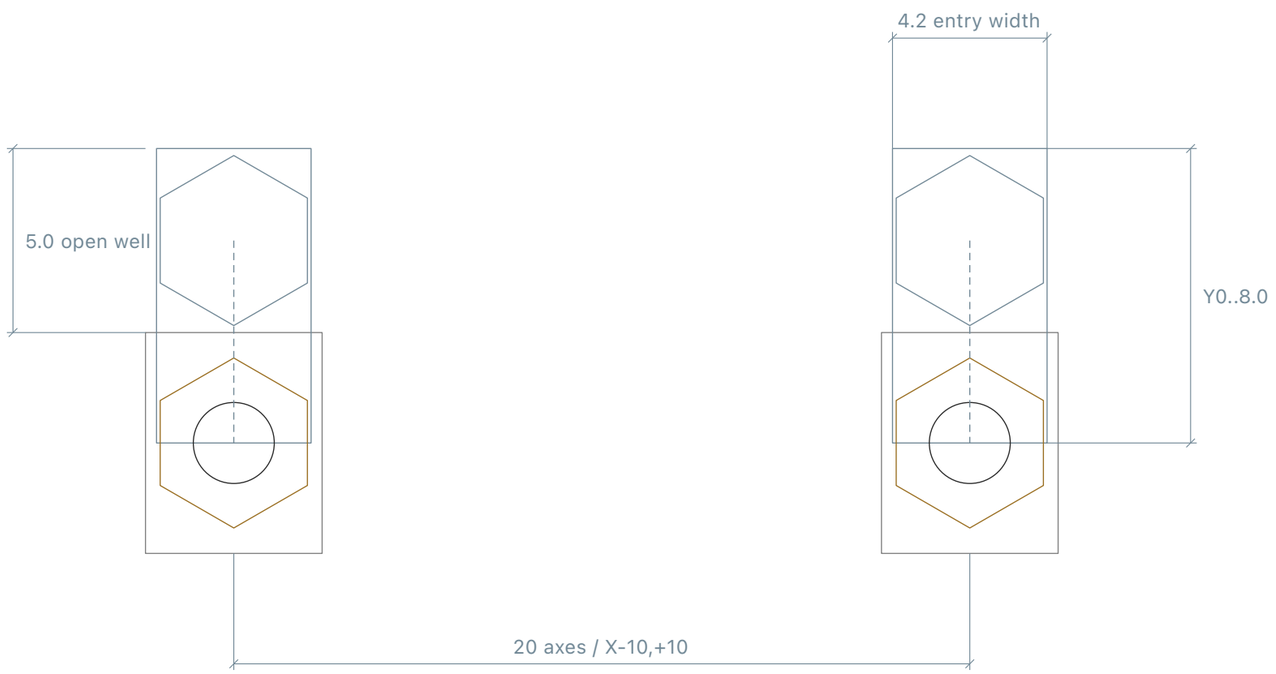
JOG WHEEL / F12-D05

15 P03 retained approved D04 PCB nut top-loading access

DRAWING NOT APPROVED | DRAFT - NOMINAL PROTOTYPE ONLY - ENGINEER DRAWING APPROVAL REQUIRED

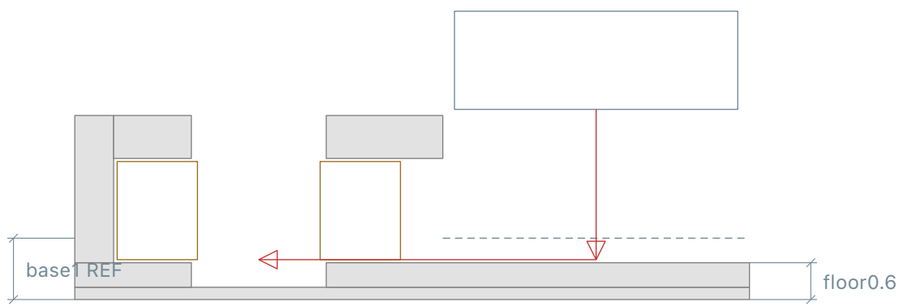
P03 BOTH ENTRIES / TOP / 6:1

Both entries identical; flats normalX; no rotation



P03 X10 SECTION / Y-Z / 10:1

RoofZ2.3 / boss topZ3; nutZ0.65..2.25



Start bottomZ3.1 ->0.65; then centreY5.5 ->0

Load before PCB/toes/cassette. Section includes unchanged blind boreD2.2 toZ0.2; nominal nut thread detail omitted. Exact bay/fastener definitions: sheets06/12. D04 SKF reference exception is nominal only; actual bearing fit remains unverified. P01/bearings unchanged.

FEATURE / mm	RETAINED APPROVED D04 ACCESS / UNCHANGED IN D05
Retained D04 cut	2 entries: X(+/-10)+[-2.1,2.1], Y0..8, Z0.6..2.3. D03 Y end3.1 only becomes8.0.
Retained geometry	Boss X widths4.8/Y+/-3/Z1..3; seated nut centres(+/-10,0), bottom0.65/top2.25; PCB back1/top3; axes unchanged.
Loading pose / path	M2 AF4/H1.6, flats normalX. Centre(+/-10,5.5), bottom3.1; lower vertically to0.65, then slideY5.5->0; no rotation.
Well clearance	Open wellY3..8: length5. Nut half-Y=4/sqrt(3)=2.309401077. Each Y margin2.5-4/sqrt(3)=0.190598923.
Inherited slot margins	X: (4.2-4)/2=0.1 each side. Z: lower0.65-0.6=0.05; upper2.3-2.25=0.05. Start bottom3.1 clears boss3 by0.1.
Floors / limitations	Relief floor0.6 retained; blind screw bore bottom0.2 retained. No strength, printability or real nut/PCB fit certification.
Coverage / authority	Approved D04 access retained unchanged. Sheets01..14 retain complete part/hardware/interface coverage; D05 changes ONLY P06 (sheet16).